

True or false?

$(9 + 4) \times 2$ is 2 more than $9 + 4$.

A. True

B. False

Which expression is 4 times as large as $125 - 25$?

A. $(125 - 25) \div 4$

B. $4 \times (125 - 25)$

C. $125 - 25 \times 4$

D. $4 + (125 - 25)$

Without calculating, how many times as large is
 $10 \times (63 + 7)$ as $63 + 7$?

Times as large: _____

SET 43 • 6.EE.A.2.b

FORWARD 1

Combine like terms with rational coefficients • Grade 7 • Expressions & Equations • 7.EE.A.1

Simplify.

$$4x + 3 - 2x + 5$$

Simplified: _____

Simplify.

$$0.5y + 1.25y$$

Simplified: _____

Which pair shows like terms?

A. $3a$ and $-7a$

B. x^2 and x

C. $3a$ and $3b$

D. $4x$ and 4

Simplify.

$$-3(x - 2) + 5x$$

Simplified: _____

True or false?

$$\frac{1}{2}x + \frac{1}{4}x = \frac{3}{4}x$$

A. True

B. False

SET 43 • 6.EE.A.2.b

FORWARD 2

Interpret the parts of a linear function • Grade 8 • Functions • 8.F.B.4

The cost to bowl is $C = 4.5n + 12$, where n is the number of games.

What does 12 represent?

Meaning: _____

In $y = -3x + 7$, what is the rate of change?

A. x

B. 3

C. -3

D. 7

The gas left in a car's tank is $G = 14 - 0.04m$, where m is miles driven.

What is the initial value?

Initial value: _____

A car's distance from home is $d = 60t + 25$, where t is hours.

What does 60 represent?

- A.** The speed in miles per hour
- B.** The starting distance
- C.** The total distance
- D.** The total time

True or false?

In $y = 8x$, the initial value is 0.

A. True

B. False

Evaluate expressions and formulas for given values

MAIN	Grade 6 main question	6.EE.A.2.c
BACKWARD 1	Whole-number exponents	6.EE.A.1
BACKWARD 2	Order of operations with grouping symbols	5.OA.A.1
BACKWARD 3	A letter stands for a number	3.OA.D.8
BACKWARD 4	Multiplying fractions	5.NF.B.4.a
FORWARD 1	Evaluate expressions with rational numbers	7.NS.A.3
FORWARD 2	Volume formulas for cylinders, cones, and spheres	8.G.C.9

SET 44 • 6.EE.A.2.c

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Evaluate $3x^2 + 2$ when $x = 4$.

Value: _____

The volume of a cube is $V = s^3$.

Find V when $s = \frac{2}{3}$ foot.

$V =$ _____

The perimeter of a rectangle is $P = 2l + 2w$.

Find P when $l = 7.5$ cm and $w = 4$ cm.

$P =$ _____

Evaluate $2(a + b)$ when $a = 3.5$ and $b = 4$.

A. 7.5

B. 9.5

C. 11.5

D. 15

Evaluate $\frac{x}{4} + 5y$ when $x = 12$ and $y = 2$.

Value: _____

SET 44 • 6.EE.A.2.c

BACKWARD 1

Whole-number exponents • Grade 6 • Expressions & Equations • 6.EE.A.1

Evaluate.

$$4^2$$

Value: _____

Evaluate.

$$\left(\frac{1}{2}\right)^3$$

Value: _____

True or false?

$$3^2 = 6$$

A. True

B. False

$$2^5 = ?$$

A. 10

B. 16

C. 25

D. 32

Evaluate.

$$5^2 + 1$$

Value: _____

SET 44 • 6.EE.A.2.c

BACKWARD 2

Order of operations with grouping symbols • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$3 \times (16 + 2)$$

Value: _____

Evaluate.

$$2 \times (6 + 4)$$

Value: _____

True or false?

$$(3 + 4) \times 2 = 11$$

A. True

B. False

Evaluate.

$$(12 \div 4) + 10$$

A. 1

B. 6

C. 13

D. 22

Evaluate.

$$[2 \times (5 + 3)] - 4$$

Value: _____

SET 44 • 6.EE.A.2.c

BACKWARD 3

A letter stands for a number • Grade 3 • Operations & Algebraic Thinking •
3.OA.D.8

If $n = 6$, what is $n + 9$?

Value: _____

True or false?

If $a = 4$, then $a \times 5 = 20$.

A. True

B. False

If $k = 7$, what is $3 \times k$?

A. 10

B. 21

C. 37

D. 73

If $m = 20$, what is $m - 8$?

Value: _____

If $t = 3$, what is $4 \times t + 2$?

Value: _____

SET 44 • 6.EE.A.2.c

BACKWARD 4

Multiplying fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$$\frac{2}{3} \times \frac{2}{3} \times \frac{2}{3}$$

Product: _____

Multiply.

$$6 \times \frac{1}{3}$$

Product: _____

True or false?

$$\frac{2}{3} \times \frac{2}{3} = \frac{4}{6}$$

A. True

B. False

$$\frac{3}{4} \times 8 = ?$$

A. $\frac{3}{32}$

B. 2

C. 6

D. 24

Multiply.

$$\frac{1}{2} \times 12$$

Product: _____

SET 44 • 6.EE.A.2.c

FORWARD 1

Evaluate expressions with rational numbers • Grade 7 • The Number System •
7.NS.A.3

Use $C = \frac{5}{9}(F - 32)$ to find C when $F = -4$.

$C =$ _____

Evaluate $-2x + 5$ when $x = -3.5$.

Value: _____

Evaluate $a^2 - b$ when $a = -3$ and $b = 4$.

A. -13

B. -5

C. 5

D. 13

Evaluate $\frac{(x+y)}{2}$ when $x = -6.5$ and $y = 2.5$.

Value: _____

True or false?

When $x = -2$, the value of $x^3 + 1$ is -7 .

A. True

B. False

SET 44 • 6.EE.A.2.c

FORWARD 2

Volume formulas for cylinders, cones, and spheres • Grade 8 • Geometry •
8.G.C.9

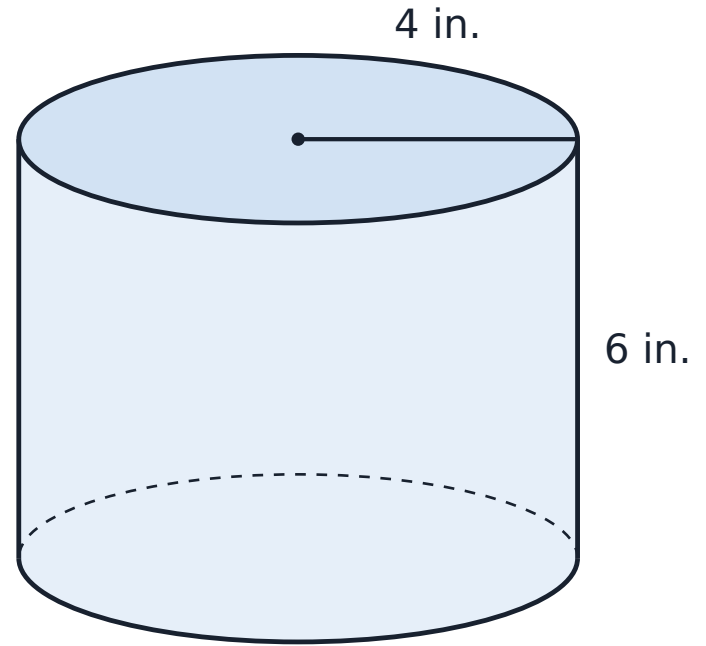
A cylinder has a radius of 3 cm and a height of 10 cm.
Find its volume. Use 3.14 for π .

Volume: _____

Find the volume of the cylinder.

Use 3.14 for π .

Volume: _____

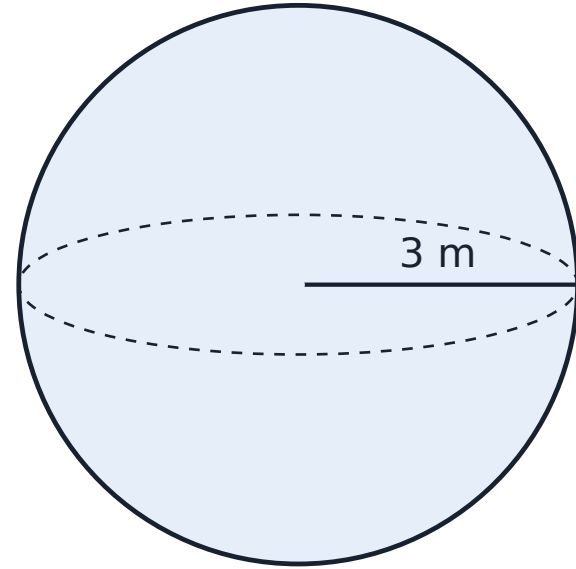


A cone has a radius of 3 units and a height of 4 units.
What is its volume in terms of π ?

- A.** 36π
- B.** 48π
- C.** 4π
- D.** 12π

Find the volume of the sphere
in terms of π .

Volume: _____



True or false?

A cylinder with a radius of 2 and a height of 5 has a volume of 20π .

A. True

B. False

Use the distributive property to expand expressions

MAIN	Grade 6 main question	6.EE.A.3
BACKWARD 1	Distributive property with numbers	3.OA.B.5
BACKWARD 2	Equal groups	3.OA.A.1
BACKWARD 3	Multiplying a decimal by a whole number	5.NBT.B.7
FORWARD 1	Expand linear expressions with rational coefficients	7.EE.A.1
FORWARD 2	Solve equations by expanding and collecting like terms	8.EE.C.7.b

SET 45 • 6.EE.A.3

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.3

Expand.

$$3(4 + x)$$

Expression: _____

Expand.

$$5(3a - 2)$$

Expression: _____

Which expression is equivalent to $7(2m + 5)$?

- A.** $9m + 35$
- B.** $2m + 35$
- C.** $14m + 5$
- D.** $14m + 35$

Expand.

$$2(x + 3y + 4)$$

Expression: _____

Expand.

$$4(0.5y + 3)$$

Expression: _____

SET 45 • 6.EE.A.3

BACKWARD 1

Distributive property with numbers • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$6 \times (4 + 3) = 6 \times 4 + 6 \times 3$$

A. True

B. False

Fill in the blank.

$$5 \times (2 + 7) = 5 \times 2 + 5 \times \underline{\quad}$$

Blank: _____

Which is equal to 3×14 ?

A. $30 + 4 + 3$

B. $3 \times 10 + 4$

C. $3 \times 10 + 3 \times 4$

D. $3 + 10 \times 4$

True or false?

$$7 \times 12 = 7 \times 10 + 2$$

A. True

B. False

Fill in the blank.

$$8 \times (5 + 1) = 8 \times 5 + \underline{\hspace{2cm}}$$

Blank: _____

SET 45 • 6.EE.A.3

BACKWARD 2

Equal groups • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

True or false?

$3 \times (4 + 2)$ means 3 groups of $(4 + 2)$.

A. True

B. False

Which means 4 groups of $(5 + 1)$?

A. $4 + (5 + 1)$

B. $4 \times (5 + 1)$

C. $4 \times 5 + 1$

D. $(4 + 5) \times 1$

How many groups of $(2 + 6)$ are in $5 \times (2 + 6)$?

Groups: _____

True or false?

$$2 \times (3 + 4) = (3 + 4) + (3 + 4)$$

A. True

B. False

Write 6×3 as a sum of equal groups.

Sum: _____

SET 45 • 6.EE.A.3

BACKWARD 3

Multiplying a decimal by a whole number • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Multiply.

$$4 \times 0.5$$

Product: _____

Multiply.

$$3 \times 1.2$$

Product: _____

True or false?

$$6 \times 0.25 = 1.5$$

A. True

B. False

$$5 \times 0.4 = ?$$

A. 0.2

B. 2

C. 5.4

D. 20

Multiply.

$$8 \times 0.75$$

Product: _____

SET 45 • 6.EE.A.3

FORWARD 1

Expand linear expressions with rational coefficients • Grade 7 • Expressions & Equations • 7.EE.A.1

Expand.

$$\frac{1}{2}(8x - 6)$$

Expression: _____

Expand.

$$-4(2y - 3)$$

Expression: _____

Which is equivalent to $0.5(6a + 10)$?

- A.** $3a + 5$
- B.** $6a + 5$
- C.** $3.5a + 5$
- D.** $3a + 10$

Expand.

$$-\frac{2}{3}(9m + 3)$$

Expression: _____

True or false?

$$-2(x - 7) = -2x + 14$$

A. True

B. False

SET 45 • 6.EE.A.3

FORWARD 2

Solve equations by expanding and collecting like terms • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$$3(x + 2) = 5x - 4$$

x = _____

Solve for x.

$$2(4x - 1) - 3x = 18$$

x = _____

Solve for x.

$$-2(x - 5) = 3(x + 5)$$

A. -5

B. -1

C. 1

D. 5

Solve for x.

$$6(x - 1) = 4(x + 2)$$

x = _____

Solve for x.

$$5(2x + 1) = 3(3x + 4)$$

x = _____

Use the distributive property to factor expressions

MAIN	Grade 6 main question	6.EE.A.3
BACKWARD 1	Greatest common factor	6.NS.B.4
BACKWARD 2	Distributive property with numbers	3.OA.B.5
BACKWARD 3	Factors	4.OA.B.4
FORWARD 1	Factor linear expressions with rational coefficients	7.EE.A.1
FORWARD 2	Use factoring to decide how many solutions an equation has	8.EE.C.7.a

SET 46 • 6.EE.A.3

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.3

Factor using the greatest common factor.

$$24x + 16$$

Expression: _____

Factor using the greatest common factor.

$$18y + 30$$

Expression: _____

Which shows $15a + 25b$ factored using the greatest common factor?

- A.** $15(a + 10b)$
- B.** $5(3a + 5b)$
- C.** $3(5a + 25b)$
- D.** $5(3a + 25b)$

Factor using the greatest common factor.

$$36m + 27n$$

Expression: _____

Factor using the greatest common factor.

$$14x + 21$$

Expression: _____

SET 46 • 6.EE.A.3

BACKWARD 1

Greatest common factor • Grade 6 • The Number System • 6.NS.B.4

Find the greatest common factor of 24 and 16.

GCF: _____

Find the greatest common factor of 18 and 30.

GCF: _____

True or false?

The greatest common factor of 15 and 25 is 5.

- A. True
- B. False

What is the greatest common factor of 36 and 27?

- A.** 3
- B.** 6
- C.** 9
- D.** 12

Find the greatest common factor of 14 and 21.

GCF: _____

SET 46 • 6.EE.A.3

BACKWARD 2

Distributive property with numbers • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$8 \times 3 + 8 \times 2 = 8 \times (3 + 2)$$

A. True

B. False

Fill in the blank.

$$6 \times 4 + 6 \times 5 = 6 \times (4 + \underline{\quad})$$

Blank: _____

Which is equal to $7 \times 2 + 7 \times 9$?

A. $14 + 9$

B. $7 \times 2 \times 9$

C. $7 \times (2 + 9)$

D. $7 + 2 + 9$

True or false?

$$5 \times 6 + 3 \times 6 = 6 \times (5 + 3)$$

A. True

B. False

Fill in the blank.

$$9 \times 7 + 9 \times 3 = \underline{\hspace{1cm}} \times (7 + 3)$$

Blank: _____

SET 46 • 6.EE.A.3

BACKWARD 3

Factors • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List all the factors of 16.

Factors: _____

True or false?

6 is a factor of 30.

A. True

B. False

Which number is a factor of 27?

- A.** 4
- B.** 6
- C.** 9
- D.** 12

List all the factors of 21.

Factors: _____

True or false?

8 is a factor of 36.

A. True

B. False

SET 46 • 6.EE.A.3

FORWARD 1

Factor linear expressions with rational coefficients • Grade 7 • Expressions & Equations • 7.EE.A.1

Factor out 2.5.

$$2.5x + 10$$

Expression: _____

Factor out -4 .

$$-4x + 12$$

Expression: _____

Which is equivalent to $\frac{1}{2}x + 3$?

A. $\frac{1}{2}(x + \frac{3}{2})$

B. $\frac{1}{2}(x + 3)$

C. $\frac{1}{2}(x + 6)$

D. $2(x + 6)$

Factor out 0.6.

$$0.6y - 1.8$$

Expression: _____

True or false?

$$-5a - 15 = -5(a + 3)$$

A. True

B. False

SET 46 • 6.EE.A.3

FORWARD 2

Use factoring to decide how many solutions an equation has • Grade 8 •
Expressions & Equations • 8.EE.C.7.a

Factor the left side to decide. How many solutions does the equation have?

$$6x + 9 = 3(2x + 3)$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

How many solutions does the equation have?

$$12x + 8 = 4(3x + 1)$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

How many solutions does the equation have?

$$10x - 15 = 5(2x - 3)$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

How many solutions does the equation have?

$$8x + 4 = 4(x + 1)$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

What number goes in the box so that the equation has infinitely many solutions?

$$14x + 21 = 7(2x + \square)$$

$\square =$ _____

Combine like terms to write equivalent expressions

MAIN	Grade 6 main question	6.EE.A.3
BACKWARD 1	Repeated addition as multiplication	3.OA.A.1
BACKWARD 2	Changing the order and grouping of addends	3.OA.B.5
FORWARD 1	Combine like terms with rational coefficients	7.EE.A.1
FORWARD 2	Solve equations by collecting like terms	8.EE.C.7.b

SET 47 • 6.EE.A.3

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.3

Write an equivalent expression with one term.

$$y + y + y + y$$

Expression: _____

Simplify.

$$5x + 3x$$

Expression: _____

Which expression is equivalent to $7a + 2 + 3a$?

A. $7a + 5a$

B. $10a + 5$

C. $10a + 2$

D. $12a$

Simplify.

$$4m + 6 + 2m + 1$$

Expression: _____

Simplify.

$$9n - 4n + 3$$

Expression: _____

SET 47 • 6.EE.A.3

BACKWARD 1

Repeated addition as multiplication • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

True or false?

$$4 + 4 + 4 = 3 \times 4$$

A. True

B. False

Write $7 + 7 + 7 + 7$ as a multiplication expression.

Expression: _____

What does 4×9 mean?

- A.** 4 groups of 9
- B.** 9 minus 4
- C.** 4 more than 9
- D.** The number 49

True or false?

$$2 + 2 + 2 + 2 + 2 = 2 \times 2$$

A. True

B. False

Write $9 + 9$ as a multiplication expression.

Expression: _____

SET 47 • 6.EE.A.3

BACKWARD 2

Changing the order and grouping of addends • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$4 + 6 + 2 = 4 + 2 + 6$$

A. True

B. False

Fill in the blank.

$$7 + 5 + 3 = 7 + 3 + \underline{\quad}$$

Blank: _____

Which is equal to $(8 + 2) + 5$?

A. $8 + 2 \times 5$

B. $8 \times (2 + 5)$

C. $(8 - 2) + 5$

D. $8 + (2 + 5)$

True or false?

$$9 + 1 + 9 = 9 + 9 + 1$$

A. True

B. False

Fill in the blank.

$$6 + 3 + 4 = 6 + 4 + \underline{\quad}$$

Blank: _____

SET 47 • 6.EE.A.3

FORWARD 1

Combine like terms with rational coefficients • Grade 7 • Expressions & Equations • 7.EE.A.1

Simplify.

$$2.25y + 0.75y$$

Expression: _____

Simplify.

$$\frac{1}{2}x + \frac{1}{4}x$$

Expression: _____

Simplify.

$$6x - 2 + 3x + 7$$

Expression: _____

Simplify.

$$-3a + 7a - 2$$

A. $4a - 2$

B. $4a + 2$

C. $2a$

D. $-10a - 2$

True or false?

$$2.5\text{m} - \text{m} = 1.5\text{m}$$

A. True

B. False

SET 47 • 6.EE.A.3

FORWARD 2

Solve equations by collecting like terms • Grade 8 • Expressions & Equations •
8.EE.C.7.b

Solve for x.

$$3x + 5x - 4 = 20$$

x = _____

Solve for y .

$$7y - 2y + 3 = 2y + 18$$

$y =$ _____

Solve for a.

$$4a + 6 - a = 2a + 10$$

A. -4

B. $\frac{16}{5}$

C. 4

D. 16

Solve for n.

$$0.5n + 1.5n - 3 = n + 5$$

n = _____

Solve for k.

$$6k - 2k + 1 = 3k - 4$$

k = _____

Identify when two expressions are equivalent

MAIN	Grade 6 main question	6.EE.A.4
BACKWARD 1	Commutative and associative properties	3.OA.B.5
BACKWARD 2	Evaluate an expression for a given value	6.EE.A.2.c
BACKWARD 3	Equal groups	3.OA.A.1
FORWARD 1	Rewrite expressions to show how quantities are related	7.EE.A.2
FORWARD 2	One, none, or infinitely many solutions	8.EE.C.7.a

SET 48 • 6.EE.A.4

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.4

True or false?

$m + m + m + m$ and $4m$ are equivalent expressions.

A. True

B. False

Which expression is equivalent to $2(x + 4)$?

A. $x + 8$

B. $2x + 4$

C. $2x + 6$

D. $2x + 8$

True or false?

$3(x + 2)$ and $3x + 2$ are equivalent expressions.

A. True

B. False

Which expression is equivalent to $6a + 4a$?

A. $6a + 4$

B. $10a$

C. $24a$

D. $10a^2$

Which pair of expressions is equivalent?

A. $5(a + 3)$ and $5a + 15$

B. $5(a + 3)$ and $a + 15$

C. $5(a + 3)$ and $5a + 3$

D. $5(a + 3)$ and $8a$

SET 48 • 6.EE.A.4

BACKWARD 1

Commutative and associative properties • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$8 + 5 = 5 + 8$$

A. True

B. False

True or false?

$$(2 \times 3) \times 4 = 2 \times (3 \times 4)$$

A. True

B. False

Which equation shows the commutative property of multiplication?

A. $6 \times 7 = 42$

B. $6 \times 7 = 7 \times 6$

C. $6 \times (7 + 1) = 6 \times 7 + 6$

D. $6 + 0 = 6$

True or false?

$$9 - 4 = 4 - 9$$

A. True

B. False

Fill in the blank.

$$3 \times 8 = 8 \times \underline{\quad}$$

Blank: _____

SET 48 • 6.EE.A.4

BACKWARD 2

Evaluate an expression for a given value • Grade 6 • Expressions & Equations •
6.EE.A.2.c

Evaluate $4x + 2x$ when $x = 3$.

Value: _____

Evaluate $3(x + 2)$ when $x = 5$.

Value: _____

True or false?

When $x = 2$, the value of $3x + 2$ is 8.

A. True

B. False

Evaluate $2(x + 4)$ when $x = 1$.

- A.** 6
- B.** 9
- C.** 10
- D.** 12

Evaluate $5a + 15$ when $a = 2$.

Value: _____

SET 48 • 6.EE.A.4

BACKWARD 3

Equal groups • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

True or false?

$$6 + 6 + 6 + 6 = 4 \times 6$$

A. True

B. False

Which expression means 3 groups of 8?

A. $3 + 8$

B. 38

C. $8 - 3$

D. 3×8

Write $5 + 5 + 5$ as a multiplication expression.

Expression: _____

True or false?

$$7 \times 2 = 7 + 7$$

A. True

B. False

How many groups of 4 are in $4 + 4 + 4 + 4 + 4$?

Groups: _____

SET 48 • 6.EE.A.4

FORWARD 1

Rewrite expressions to show how quantities are related • Grade 7 •
Expressions & Equations • 7.EE.A.2

A price c plus an 8% tax can be written as $c + 0.08c$.

Which expression is equivalent?

- A.** $0.08c$
- B.** $1.08c$
- C.** $1.8c$
- D.** $8.08c$

True or false?

$$a - 0.2a = 0.8a$$

A. True

B. False

Write an expression with one term that is equivalent to $n - 0.25n$.

Expression: _____

Which expression shows the price p after a 30% discount?

- A.** $0.3p$
- B.** $0.7p$
- C.** $1.3p$
- D.** $p - 30$

The perimeter of a square is $s + s + s + s$.

Write an equivalent expression with one term.

Expression: _____

SET 48 • 6.EE.A.4

FORWARD 2

One, none, or infinitely many solutions • Grade 8 • Expressions & Equations •
8.EE.C.7.a

True or false?

The equation $2(x + 3) = 2x + 6$ has infinitely many solutions.

A. True

B. False

How many solutions does the equation have?

$$5x - 3 = 5x + 1$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

How many solutions does the equation have?

$$3(2x + 4) = 6x + 12$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

Write a number in the box so the equation has no solution.

$$4x + 7 = 4x + \square$$

$$\square = \underline{\hspace{10cm}}$$

Which equation has exactly one solution?

A. $3x + 1 = 2x + 1$

B. $3x + 1 = 3x + 2$

C. $3x + 1 = 3x + 1$

D. $3(x + 1) = 3x + 3$

Use substitution to decide whether a number makes an equation or inequality true

MAIN	Grade 6 main question	6.EE.B.5
BACKWARD 1	Evaluate an expression	6.EE.A.2.c
BACKWARD 2	Comparing numbers	2.NBT.A.4
BACKWARD 3	Meaning of the equal sign	1.OA.D.7
FORWARD 1	Solve equations of the form $px + q = r$ and $p(x + q) = r$	7.EE.B.4.a
FORWARD 2	Solutions of a system of two linear equations	8.EE.C.8.a

SET 49 • 6.EE.B.5

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.5

Which value from the set $\{2, 4, 6, 8\}$ makes $3x + 1 = 19$ true?

A. 2

B. 4

C. 6

D. 8

True or false?

$x = 5$ is a solution of $2x - 3 > 8$.

A. True

B. False

Which numbers from the set $\{1, 3, 5, 7\}$ make $x + 4 < 10$ true?

Numbers: _____

Is $y = 2.5$ a solution of $4y = 10$?

A. Yes

B. No

Which value from the set $\{12, 15, 18\}$ makes $n \div 3 = 6$ true?

$n =$ _____

SET 49 • 6.EE.B.5

BACKWARD 1

Evaluate an expression • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Evaluate $3x + 1$ when $x = 6$.

Value: _____

Evaluate $2x - 3$ when $x = 5$.

Value: _____

True or false?

When $y = 4$, the value of $5y$ is 20.

A. True

B. False

Evaluate $x + 4$ when $x = 7$.

- A.** 3
- B.** 11
- C.** 28
- D.** 47

Evaluate $n \div 3$ when $n = 15$.

Value: _____

SET 49 • 6.EE.B.5

BACKWARD 2

Comparing numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

Write $>$, $<$, or $=$ to compare.

19 21

Symbol: _____

True or false?

$$7 > 8$$

A. True

B. False

Which statement is true?

A. $9 > 6$

B. $10 < 5$

C. $11 < 10$

D. $6 > 9$

True or false?

$$24 = 24$$

A. True

B. False

Write $>$, $<$, or $=$ to compare.

105 150

Symbol: _____

SET 49 • 6.EE.B.5

BACKWARD 3

Meaning of the equal sign • Grade 1 • Operations & Algebraic Thinking •
1.OA.D.7

True or false?

$$4 + 5 = 9 + 0$$

A. True

B. False

True or false?

$$12 = 15 - 3$$

A. True

B. False

True or false?

$$3 + 6 = 5 + 3$$

A. True

B. False

Which equation is true?

A. $4 + 3 = 5 + 3$

B. $4 + 3 = 5 + 2$

C. $7 = 4 - 3$

D. $4 + 3 = 8$

True or false?

$$10 - 2 = 7$$

A. True

B. False

SET 49 • 6.EE.B.5

FORWARD 1

Solve equations of the form $px + q = r$ and $p(x + q) = r$ • Grade 7 •
Expressions & Equations • 7.EE.B.4.a

Solve for x.

$$3x + 7 = 22$$

x = _____

Solve for x.

$$4(x - 2) = 20$$

x = _____

Solve for x.

$$0.5x + 6 = 10$$

A. 2

B. 8

C. 20

D. 32

Solve for x.

$$\frac{2}{3}x - 4 = 10$$

x = _____

A movie ticket costs \$12. Popcorn costs p dollars. Two tickets and one popcorn cost \$31.

What is p ?

$p =$ _____

SET 49 • 6.EE.B.5

FORWARD 2

Solutions of a system of two linear equations • Grade 8 • Expressions & Equations • 8.EE.C.8.a

True or false?

$(2, 3)$ is a solution of both $y = x + 1$ and $y = 2x - 1$.

A. True

B. False

Which point is a solution of both $y = 2x$ and $y = x + 3$?

A. (0, 3)

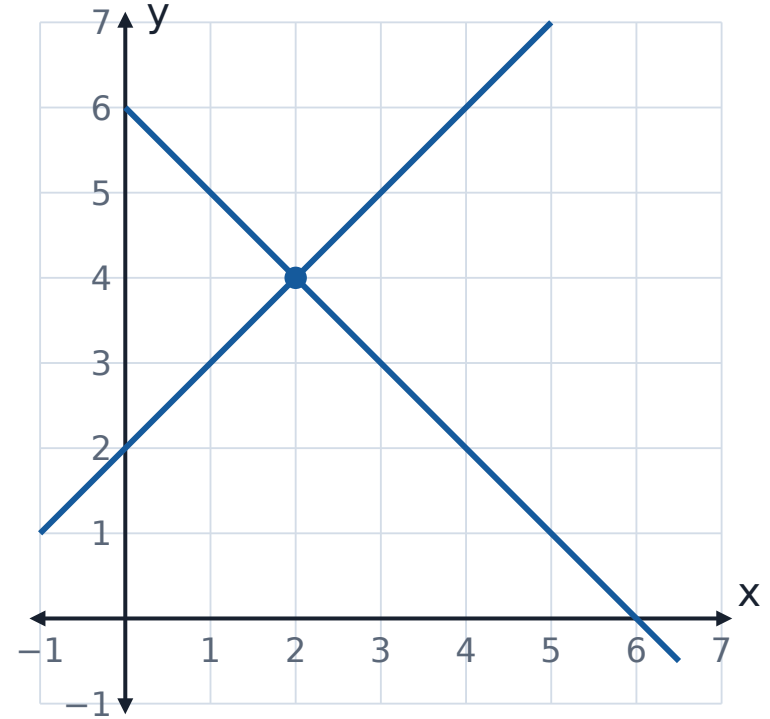
B. (2, 5)

C. (1, 2)

D. (3, 6)

What is the solution of the system shown by the graph?

Solution: _____



True or false?

(1, 4) is a solution of both $y = 3x + 1$ and $y = -x + 5$.

A. True

B. False

Is $(0, 2)$ a solution of both $y = 2x + 2$ and $y = x - 2$?

A. Yes

B. No

Use variables to write expressions for real-world problems

MAIN	Grade 6 main question	6.EE.B.6
BACKWARD 1	Word problems with an unknown number	3.OA.A.3
BACKWARD 2	Multiplicative comparison word problems	4.OA.A.2
BACKWARD 3	Number patterns that follow a rule	4.OA.C.5
FORWARD 1	Write and solve equations for word problems	7.EE.B.4.a
FORWARD 2	Model a linear relationship with a function	8.F.B.4

SET 50 • 6.EE.B.6

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.6

Tickets cost \$8 each.

Write an expression for the cost of t tickets.

Expression: _____

Kim is k years old. Sam is 3 years older than Kim.

Write an expression for Sam's age.

Expression: _____

A pizza is cut into s slices. 4 friends share the slices equally.

Which expression shows how many slices each friend gets?

A. $4s$

B. $s - 4$

C. $\frac{s}{4}$

D. $s + 4$

The length of a rectangle is 5 inches more than its width w .

Write an expression for the length.

Expression: _____

A gym charges \$20 to join plus \$15 per month.

Write an expression for the total cost after m months.

Expression: _____

SET 50 • 6.EE.B.6

BACKWARD 1

Word problems with an unknown number • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

There are 6 bags with the same number of apples.
There are 42 apples in all.

How many apples are in each bag?

Apples: _____

There are 5 boxes with $\square$ pencils in each box. There are 35 pencils in all.

Which equation matches?

A. $35 \times 5 = \square$

B. $\square - 5 = 35$

C. $5 + \square = 35$

D. $5 \times \square = 35$

True or false?

If $4 \times \square = 28$, then $\square = 7$.

A. True

B. False

There are 8 rows with $\square$ chairs in each row. There are 56 chairs in all.

What number is $\square$?

$\square =$ _____

Jan split 45 stickers equally among 9 friends.
How many stickers did each friend get?

Stickers: _____

SET 50 • 6.EE.B.6

BACKWARD 2

Multiplicative comparison word problems • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2

A red ribbon is 7 inches long. A blue ribbon is 3 times as long.

How long is the blue ribbon?

Length: _____

Ben has 6 books. Ali has 4 times as many books as Ben.
Which equation finds b , the number of books Ali has?

A. $b = 4 \times 6$

B. $b = 4 + 6$

C. $b = 6 \div 4$

D. $b = 6 - 4$

True or false?

A cat weighs 5 pounds. A dog weighs 8 times as much. The dog weighs 40 pounds.

- A. True
- B. False

A bush is 4 feet tall. A tree is 5 times as tall.

How tall is the tree?

Height: _____

Sara ran 24 miles. That is 3 times as far as Tom ran.

How far did Tom run?

Miles: _____

SET 50 • 6.EE.B.6

BACKWARD 3

Number patterns that follow a rule • Grade 4 • Operations & Algebraic Thinking
• 4.OA.C.5

The rule is "add 4." The pattern starts at 3.
Write the first five numbers.

Numbers: _____

The rule is "multiply by 2." The pattern starts at 1.
What is the 5th number?

- A.** 8
- B.** 10
- C.** 16
- D.** 32

True or false?

The rule "add 5," starting at 5, gives 5, 10, 15, 20.

A. True

B. False

What is the rule for this pattern?

7, 11, 15, 19, ...

Rule: _____

The rule is "add 6." The pattern starts at 2.

What is the 4th number?

Number: _____

SET 50 • 6.EE.B.6

FORWARD 1

Write and solve equations for word problems • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Sam has \$60. He spends \$8 each week.

Write and solve an equation to find when he will have \$20 left.

Weeks: _____

A rectangle has a perimeter of 50 cm and a length of 15 cm.

Which equation can be used to find its width w ?

A. $2(15 + w) = 50$

B. $2(15w) = 50$

C. $15 + 2w = 50$

D. $15 + w = 50$

A taxi charges \$3 plus \$2 per mile. The fare was \$17.

How many miles was the ride?

Miles: _____

You buy 4 notebooks and a \$6 backpack. You spend \$18 in all.

How much does each notebook cost?

Cost: _____

True or false?

"Five less than twice a number is 17" can be written as $2n - 5 = 17$.

A. True

B. False

SET 50 • 6.EE.B.6

FORWARD 2

Model a linear relationship with a function • Grade 8 • Functions • 8.F.B.4

A plant is 4 cm tall. It grows 1.5 cm each week.
Write a function for its height h after w weeks.

$h =$ _____

Which situation matches $y = 2x + 10$, where y is dollars in savings after x weeks?

- A.** Save \$12 each week.
- B.** Start with \$10 and spend \$2 each week.
- C.** Start with \$10 and save \$2 each week.
- D.** Start with \$2 and save \$10 each week.

The table shows savings over time.

Write a function for the savings y after x weeks.

Weeks (x)	0	1	2
Savings (y)	30	45	60

$y =$ _____

A rental truck costs \$50 plus \$0.25 per mile.

Write a function for the cost C of driving m miles.

$C =$ _____

True or false?

A line through $(0, 5)$ and $(2, 9)$ has an initial value of 5 and a rate of change of 2.

A. True

B. False

Solve one-step equations of the form

$$x + p = q$$

MAIN	Grade 6 main question	6.EE.B.7
BACKWARD 1	Subtraction as an unknown-addend problem	1.OA.B.4
BACKWARD 2	Adding and subtracting decimals	5.NBT.B.7
BACKWARD 3	Adding and subtracting fractions and mixed numbers	5.NF.A.1
FORWARD 1	Solve two-step equations	7.EE.B.4.a
FORWARD 2	Solve equations with variables on both sides	8.EE.C.7.b

SET 51 • 6.EE.B.7

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

$$x + 7.5 = 12$$

x = _____

Solve for n.

$$n - \frac{3}{4} = 5\frac{1}{2}$$

A. $4\frac{3}{4}$

B. $5\frac{1}{4}$

C. $6\frac{1}{4}$

D. $6\frac{3}{4}$

Solve for y.

$$y + 2\frac{1}{3} = 6$$

y = _____

Solve for m.

$$18.25 + m = 30$$

m = _____

Solve for k.

$$k - 4.8 = 9.2$$

$$k = \underline{\hspace{10cm}}$$

SET 51 • 6.EE.B.7

BACKWARD 1

Subtraction as an unknown-addend problem • Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4

True or false?

$12 - 7 = \square$ can be solved by thinking
 $7 + \square = 12$.

A. True

B. False

What number goes in the box?

$$7 + \square = 15$$

$$\square = \underline{\hspace{10em}}$$

To find $16 - 9$, think $9 + \square = 16$.

What number goes in the box?

- A.** 6
- B.** 7
- C.** 8
- D.** 25

What number goes in the box?

$$\square + 8 = 13$$

$$\square = \underline{\hspace{10em}}$$

True or false?

To find $11 - 5$, you can think $5 + 6 = 11$.

A. True

B. False

SET 51 • 6.EE.B.7

BACKWARD 2

Adding and subtracting decimals • Grade 5 • Number & Operations in Base Ten
• 5.NBT.B.7

Subtract.

$$12 - 7.5$$

Difference: _____

Subtract.

$$30 - 18.25$$

Difference: _____

True or false?

$$9.2 + 4.8 = 14$$

A. True

B. False

$$6.25 + 2.8 = ?$$

A. 6.53

B. 8.33

C. 9.05

D. 9.5

Add.

$$3.7 + 5.45$$

Sum: _____

SET 51 • 6.EE.B.7

BACKWARD 3

Adding and subtracting fractions and mixed numbers • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Subtract.

$$6 - 2\frac{1}{3}$$

Difference: _____

Add.

$$5\frac{1}{2} + \frac{3}{4}$$

Sum: _____

True or false?

$$\frac{2}{3} + \frac{1}{6} = \frac{5}{6}$$

A. True

B. False

$$\frac{7}{8} - \frac{1}{4} = ?$$

A. $\frac{1}{2}$

B. $\frac{5}{8}$

C. $\frac{3}{4}$

D. $\frac{6}{4}$

Add.

$$1\frac{2}{5} + \frac{1}{2}$$

Sum: _____

SET 51 • 6.EE.B.7

FORWARD 1

Solve two-step equations • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for x.

$$3x + 8 = 29$$

x = _____

Solve for x.

$$-2x + 5 = 17$$

x = _____

Solve for x.

$$\frac{x}{4} - 3 = 2$$

A. -4

B. $\frac{5}{4}$

C. 5

D. 20

Solve for x.

$$0.6x + 1.5 = 4.5$$

x = _____

Solve for x.

$$5(x + 2) = 45$$

x = _____

SET 51 • 6.EE.B.7

FORWARD 2

Solve equations with variables on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$$7x - 4 = 3x + 12$$

x = _____

Solve for x.

$$-5x + 9 = x - 3$$

x = _____

Solve for x.

$$2.5x + 4 = 1.5x + 9$$

A. -5

B. 5

C. 6.5

D. 13

Solve for x.

$$\frac{1}{2}x + 6 = 2x - 3$$

x = _____

Solve for x.

$$3(2x - 5) = 4x + 1$$

x = _____

Solve one-step equations of the form

$$px = q$$

MAIN	Grade 6 main question	6.EE.B.7
BACKWARD 1	Division as an unknown-factor problem	3.OA.B.6
BACKWARD 2	Dividing by a fraction	6.NS.A.1
BACKWARD 3	Dividing decimals	5.NBT.B.7
FORWARD 1	Solve equations of the form $p(x + q) = r$	7.EE.B.4.a
FORWARD 2	Solve multi-step linear equations	8.EE.C.7.b

SET 52 • 6.EE.B.7

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

$$4x = 30$$

x = _____

Solve for y.

$$\frac{2}{3}y = 12$$

$$y = \underline{\hspace{10cm}}$$

Solve for m.

$$0.6m = 4.2$$

$$m = \underline{\hspace{10em}}$$

Solve for n.

$$5n = 3.5$$

A. -1.5

B. 0.7

C. 1.43

D. 17.5

Solve for k.

$$\frac{3}{4}k = \frac{9}{10}$$

k = _____

SET 52 • 6.EE.B.7

BACKWARD 1

Division as an unknown-factor problem • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6

Find $48 \div 6$ by thinking $6 \times \square = 48$.

$\square =$ _____

True or false?

$63 \div 7 = 9$ because $7 \times 9 = 63$.

A. True

B. False

Which fact helps you find $45 \div 5$?

A. $9 - 5 = 4$

B. $45 \times 5 = 225$

C. $5 \times 9 = 45$

D. $5 + 40 = 45$

What number goes in the box?

$$\square \times 6 = 42$$

$$\square = \underline{\hspace{10em}}$$

True or false?

$24 \div 4 = 5$ because $4 \times 5 = 24$.

A. True

B. False

SET 52 • 6.EE.B.7

BACKWARD 2

Dividing by a fraction • Grade 6 • The Number System • 6.NS.A.1

Divide.

$$12 \div \frac{2}{3}$$

Quotient: _____

Divide.

$$\frac{9}{10} \div \frac{3}{4}$$

Quotient: _____

True or false?

$$5 \div \frac{5}{8} = 8$$

A. True

B. False

$$\frac{9}{10} \div \frac{3}{5} = ?$$

A. $\frac{27}{50}$

B. $\frac{2}{3}$

C. $1\frac{1}{3}$

D. $1\frac{1}{2}$

Divide.

$$6 \div \frac{3}{4}$$

Quotient: _____

SET 52 • 6.EE.B.7

BACKWARD 3

Dividing decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Divide.

$$4.2 \div 0.6$$

Quotient: _____

Divide.

$$30 \div 4$$

Quotient: _____

True or false?

$$3.5 \div 5 = 0.7$$

A. True

B. False

$$2.4 \div 0.3 = ?$$

A. 0.8

B. 7

C. 8

D. 80

Divide.

$$7.2 \div 9$$

Quotient: _____

SET 52 • 6.EE.B.7

FORWARD 1

Solve equations of the form $p(x + q) = r$ • Grade 7 • Expressions & Equations •
7.EE.B.4.a

Solve for x.

$$3(x - 4) = 21$$

x = _____

Solve for x.

$$\frac{1}{2}(x + 6) = 9$$

x = _____

Solve for x.

$$3(x + 1.5) = 12$$

A. 2

B. 2.5

C. 4

D. 5.5

Solve for x.

$$-2(x - 3) = 14$$

x = _____

Solve for x.

$$0.5(x + 8) = 7$$

x = _____

SET 52 • 6.EE.B.7

FORWARD 2

Solve multi-step linear equations • Grade 8 • Expressions & Equations •
8.EE.C.7.b

Solve for x.

$$3(2x - 1) = 4x + 7$$

x = _____

Solve for x.

$$\frac{2}{3}(x + 3) = x - 1$$

x = _____

Solve for x.

$$5x - 2(x + 1) = 10$$

A. 2

B. $\frac{8}{3}$

C. 4

D. 12

Solve for x.

$$0.4(10x - 5) = 2x + 4$$

x = _____

Solve for x.

$$6x = 2(x + 8)$$

x = _____

Write and solve one-step equations for real-world problems

MAIN	Grade 6 main question	6.EE.B.7
BACKWARD 1	Word problems with an unknown factor	3.OA.A.3
BACKWARD 2	Word problems with an unknown addend	2.OA.A.1
BACKWARD 3	A letter stands for an unknown quantity	3.OA.D.8
FORWARD 1	Write and solve two-step equations for word problems	7.EE.B.4.a
FORWARD 2	Solve real-world problems with variables on both sides	8.EE.C.7.b

SET 53 • 6.EE.B.7

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.7

A pack of 6 pens costs \$4.50.

Write and solve an equation to find the cost p of one pen.

Equation: _____

$p =$ _____

After spending \$12.75, Sam has \$30.25 left.

Write and solve an equation to find how much money m Sam started with.

Equation: _____

$m =$ _____

A board is cut into 5 equal pieces. Each piece is 1.8 meters long.

Write and solve an equation to find the length L of the board.

Equation: _____

$L =$ _____

Maya is 1.35 m tall. She is 0.4 m taller than her brother.
Which equation can be used to find her brother's height b ?

A. $0.4b = 1.35$

B. $1.35 + 0.4 = b$

C. $b - 0.4 = 1.35$

D. $b + 0.4 = 1.35$

A recipe uses $\frac{3}{4}$ cup of sugar per batch. Kim used 6 cups of sugar.

Write and solve an equation to find the number of batches b .

Equation: _____

$b =$ _____

SET 53 • 6.EE.B.7

BACKWARD 1

Word problems with an unknown factor • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

4 friends share some grapes equally. Each gets 9 grapes.

How many grapes were there?

Grapes: _____

There are 7 teams with $\square$ players each. There are 56 players in all.

Which equation matches?

A. $7 \times \square = 56$

B. $56 \times 7 = \square$

C. $\square - 7 = 56$

D. $7 + \square = 56$

True or false?

If $3 \times \square = 24$, then $\square = 8$.

A. True

B. False

A teacher puts 40 books into 5 equal stacks.
How many books are in each stack?

Books: _____

Each box holds 6 eggs. There are 54 eggs.

How many boxes are there?

Boxes: _____

SET 53 • 6.EE.B.7

BACKWARD 2

Word problems with an unknown addend • Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

Lin had some stickers. She got 15 more. Now she has 42.

How many did she have at first?

Stickers: _____

A plant grew 8 cm and is now 35 cm tall.

Which equation finds the starting height h ?

A. $8h = 35$

B. $h - 8 = 35$

C. $h + 35 = 8$

D. $h + 8 = 35$

True or false?

Tom spent \$17 and has \$26 left, so he started with \$43.

- A. True
- B. False

A bus had 30 people. Some got off. Now there are 18.
How many got off?

People: _____

Ada read 24 pages on Monday and some more on Tuesday. She read 61 pages in all.

How many pages did she read on Tuesday?

Pages: _____

SET 53 • 6.EE.B.7

BACKWARD 3

A letter stands for an unknown quantity • Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8

True or false?

In $5 \times n = 40$, the letter n stands for 8.

A. True

B. False

In the equation $p + 9 = 20$, what number is p ?

$p =$ _____

Which equation matches "a number plus 6 is 15"?

A. $n + 15 = 6$

B. $n - 6 = 15$

C. $6n = 15$

D. $n + 6 = 15$

In the equation $3 \times k = 27$, what number is k ?

$k =$ _____

True or false?

In $m - 4 = 10$, the letter m stands for 6.

A. True

B. False

SET 53 • 6.EE.B.7

FORWARD 1

Write and solve two-step equations for word problems • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A gym charges a \$25 fee plus \$15 per month. Jen paid \$130.

Write and solve an equation to find the number of months m .

$m =$ _____

A plumber charges \$45 plus \$60 per hour. A bill is \$225.
How many hours did the plumber work?

Hours: _____

A class buys 12 tickets and pays a \$15 group fee. The total is \$159.

Which equation finds the cost t of one ticket?

A. $12t - 15 = 159$

B. $12t + 15 = 159$

C. $12(t + 15) = 159$

D. $15t + 12 = 159$

Mo saves \$18 per week and already has \$54. He wants \$180.

How many weeks w will it take?

$W =$ _____

True or false?

A \$6 pizza plus 3 toppings costs \$10.50. Each topping costs \$1.50.

- A. True
- B. False

SET 53 • 6.EE.B.7

FORWARD 2

Solve real-world problems with variables on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Store A charges \$20 plus \$3 per shirt. Store B charges \$5 per shirt.

Write and solve an equation to find when the costs are equal.

Shirts: _____

Ana has \$120 and spends \$8 per week. Ben has \$40 and saves \$12 per week.

After how many weeks will they have the same amount?

Weeks: _____

Taxi A charges \$4 plus \$2.50 per mile. Taxi B charges \$1 plus \$3 per mile.

For how many miles do the rides cost the same?

A. 3

B. 5

C. 6

D. 10

A candle 30 cm tall burns 1.5 cm per hour. Another candle 20 cm tall burns 0.5 cm per hour.

After how many hours are they the same height?

Hours: _____

Plan A: \$15 per month plus \$0.10 per text. Plan B: \$0.25 per text.

For how many texts do the plans cost the same?

Texts: _____

Write an inequality $x > c$ or $x < c$ for a real-world condition

MAIN	Grade 6 main question	6.EE.B.8
BACKWARD 1	Inequality symbols	1.NBT.B.3
BACKWARD 2	Meaning of "at least," "at most," "more than," and "fewer than"	6.NS.C.7.b
BACKWARD 3	Negative numbers in context	6.NS.C.5
FORWARD 1	Write and solve inequalities for real-world problems	7.EE.B.4.b
FORWARD 2	Use two linear equations to decide when one quantity is greater...	8.EE.C.8.c

SET 54 • 6.EE.B.8

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.8

Riders must be at least 48 inches tall.

Write an inequality for the height h of a rider.

Inequality: _____

A bag can hold at most 20 pounds.

Which inequality describes the weight w the bag can hold?

A. $w \geq 20$

B. $w < 20$

C. $w \leq 20$

D. $w > 20$

The temperature must stay below -3°C .

Write an inequality for the temperature t .

Inequality: _____

More than 25 people must sign up for a trip to happen.
Write an inequality for the number of people p .

Inequality: _____

Children 12 years old and younger get a discount.

Write an inequality for the age a of a child who gets the discount.

Inequality: _____

SET 54 • 6.EE.B.8

BACKWARD 1

Inequality symbols • Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3

Which symbol makes this true?

58 85

A. <

B. >

C. =

True or false?

$$30 > 29$$

A. True

B. False

Write $>$, $<$, or $=$ to compare.

64 46

Symbol: _____

Which symbol means "is greater than"?

A. $<$

B. $>$

C. $=$

True or false?

$$17 < 12$$

A. True

B. False

SET 54 • 6.EE.B.8

BACKWARD 2

Meaning of "at least," "at most," "more than," and "fewer than" • Grade 6 • The Number System • 6.NS.C.7.b

True or false?

"At least 10" includes the number 10.

A. True

B. False

What does "at most 6" mean?

- A.** 6 or more
- B.** Less than 6
- C.** More than 6
- D.** 6 or less

True or false?

"Fewer than 8" includes the number 8.

A. True

B. False

Which number is NOT "at least 12"?

- A.** 11
- B.** 12
- C.** 13
- D.** 20

True or false?

9 is "more than 7."

A. True

B. False

SET 54 • 6.EE.B.8

BACKWARD 3

Negative numbers in context • Grade 6 • The Number System • 6.NS.C.5

Write an integer for 3 degrees below zero.

Integer: _____

True or false?

-10°C is colder than -3°C .

A. True

B. False

Which number represents 15 feet below sea level?

- A.** -15
- B.** -1.5
- C.** 0
- D.** 15

Write an integer for a temperature of 7 degrees below zero.

Integer: _____

True or false?

"Below -3°C " includes -5°C .

A. True

B. False

SET 54 • 6.EE.B.8

FORWARD 1

Write and solve inequalities for real-world problems • Grade 7 • Expressions & Equations • 7.EE.B.4.b

You have \$50. Each ride costs \$6.

Write and solve an inequality to find how many rides r you can afford.

Solution: _____

A salesperson earns \$80 plus \$12 for each sale.

Write and solve an inequality for the number of sales s needed to earn more than \$200.

Solution: _____

A phone has 75% battery and uses 5% per hour. It must stay above 20%.

Which inequality gives the hours h ?

A. $75 - 5h > 20$

B. $75 + 5h > 20$

C. $75 - 5h < 20$

D. $5h - 75 > 20$

A van can carry at most 1,200 pounds. The driver weighs 180 pounds. Each box weighs 60 pounds.

Write and solve an inequality for the number of boxes b .

Solution: _____

True or false?

The solution of $3x + 2 \geq 17$ is $x \geq 5$.

A. True

B. False

FORWARD 2

Use two linear equations to decide when one quantity is greater • Grade 8 •
Expressions & Equations • 8.EE.C.8.c

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Plan A costs \$20 plus \$5 per month. Plan B costs \$35 plus \$2 per month.

After how many months does Plan A cost more than Plan B?

Months: _____

Tank A has 100 gallons and drains 6 gallons per minute.
Tank B has 40 gallons and fills 4 gallons per minute.
After how many minutes does Tank B hold more water?

Minutes: _____

Sara has \$50 and saves \$8 per week. Tom has \$90 and saves \$3 per week.

When will Sara have more money than Tom?

- A.** After 8 weeks
- B.** After 3 weeks
- C.** After 5 weeks
- D.** Never

Gym X charges \$60 plus \$4 per class. Gym Y charges \$12 per class.

For how many classes is Gym X cheaper?

Classes: _____

True or false?

$y = 3x + 2$ and $y = x + 10$ are equal when $x = 4$.
For $x > 4$, $3x + 2$ is greater.

A. True

B. False

Represent solutions of $x > c$ or $x < c$ on a number line

MAIN	Grade 6 main question	6.EE.B.8
BACKWARD 1	Locating numbers on a number line	6.NS.C.6.c
BACKWARD 2	Greater numbers are to the right	6.NS.C.7.a
BACKWARD 3	Meaning of inequality symbols	1.NBT.B.3
FORWARD 1	Solve and graph two-step inequalities	7.EE.B.4.b
FORWARD 2	Linear equations with one, no, or infinitely many solutions — near...	8.EE.C.7.a

SET 55 • 6.EE.B.8

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.8

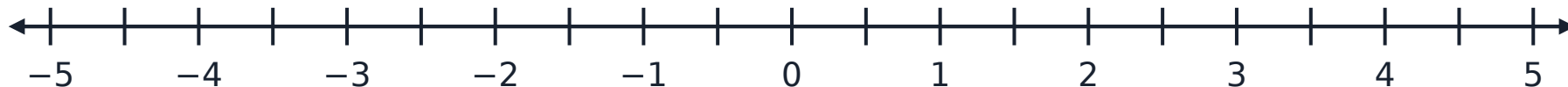
Graph the solutions of $x > 3$ on the number line.



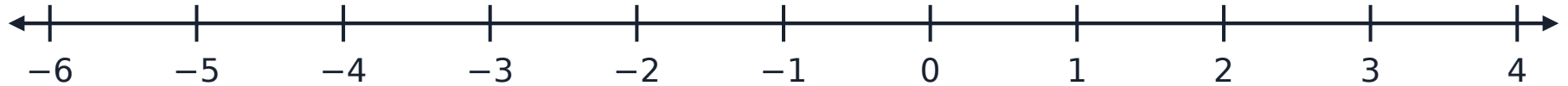
Graph the solutions of $x \leq -1$ on the number line.



Graph the solutions of $x < 2.5$ on the number line.



Graph the solutions of $x \geq -4$ on the number line.



Graph the solutions of $x > -1.5$ on the number line.



SET 55 • 6.EE.B.8

BACKWARD 1

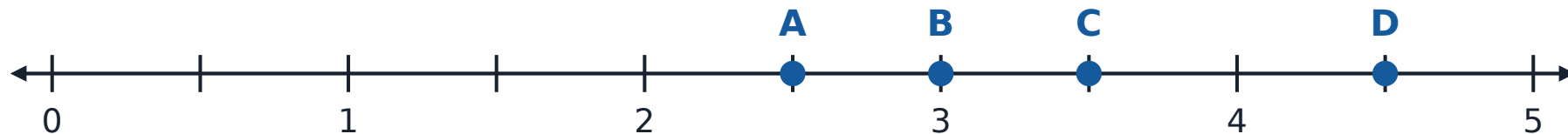
Locating numbers on a number line • Grade 6 • The Number System •
6.NS.C.6.c

What number is at point A?



A = _____

Which point is at 3.5?



A. Point A

B. Point B

C. Point C

D. Point D

True or false?

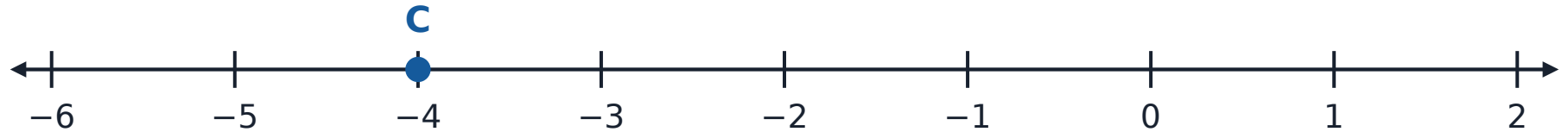
Point B is at -1.5 .



A. True

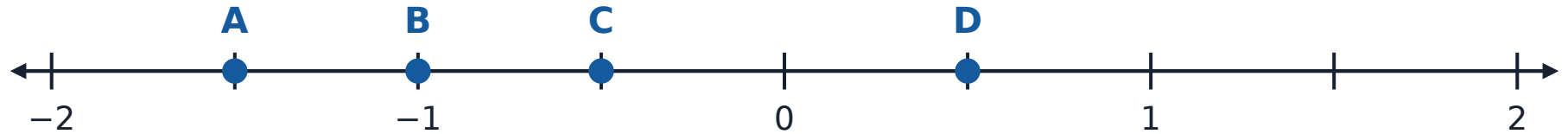
B. False

What number is at point C?



C = _____

Which point is at $-\frac{1}{2}$?



A. Point A

B. Point B

C. Point C

D. Point D

SET 55 • 6.EE.B.8

BACKWARD 2

Greater numbers are to the right • Grade 6 • The Number System • 6.NS.C.7.a

True or false?

Every number to the right of 3 on a number line is greater than 3.

- A. True
- B. False

Which numbers are less than -1 ?

- A.** Numbers to the right of -1
- B.** Numbers to the left of -1
- C.** Only -2
- D.** Only 0

True or false?

-3 is greater than -1 .

A. True

B. False

Is 2.5 to the left or the right of 2 on a number line?

Answer: _____

Which number is greater than -4 ?

A. -6

B. -5

C. -4.5

D. -3

SET 55 • 6.EE.B.8

BACKWARD 3

Meaning of inequality symbols • Grade 1 • Number & Operations in Base Ten •
1.NBT.B.3

What does the symbol $>$ mean?

- A.** Is equal to
- B.** Is greater than
- C.** Is less than

True or false?

The symbol $<$ means "is less than."

- A. True
- B. False

Which statement is true?

A. $8 < 5$

B. $8 = 5$

C. $8 > 5$

Write the symbol for "is less than."

Symbol: _____

True or false?

$$12 > 21$$

A. True

B. False

SET 55 • 6.EE.B.8

FORWARD 1

Solve and graph two-step inequalities • Grade 7 • Expressions & Equations •
7.EE.B.4.b

Solve $2x + 1 > 7$. Then graph the solutions.



Solution: _____

Solve $4x - 3 \leq 5$. Then graph the solutions.



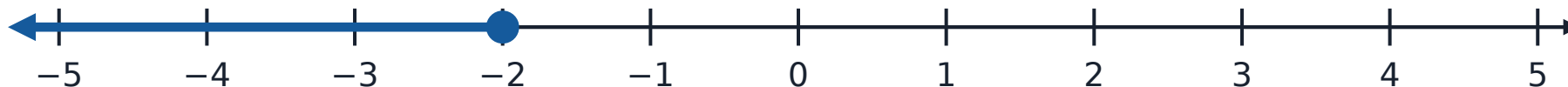
Solution: _____

Solve $-2x + 4 < 10$. Then graph the solutions.



Solution: _____

Which inequality does the graph show?



A. $x < -2$

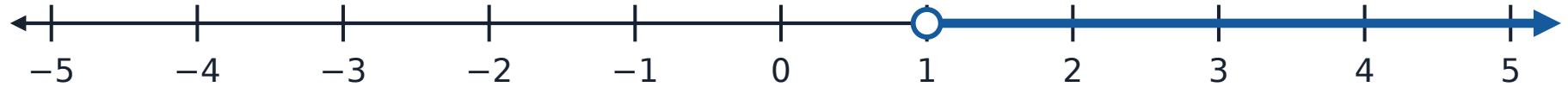
B. $x > -2$

C. $x \geq -2$

D. $x \leq -2$

True or false?

The graph shows the solutions of $3x + 2 > 5$.



A. True

B. False

FORWARD 2

Linear equations with one, no, or infinitely many solutions • Grade 8 •
Expressions & Equations • 8.EE.C.7.a

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

True or false?

$x + 3 = x + 3$ has infinitely many solutions.

A. True

B. False

How many solutions does the equation have?

$$2x + 5 = 2x - 1$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

Which equation has infinitely many solutions?

A. $4x - 1 = x$

B. $4x = 0$

C. $4(x - 1) = 4x - 4$

D. $4(x - 1) = 4x - 1$

How many solutions does the equation have?

$$3x - 7 = 3x - 7$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

How many solutions does the equation have?

$$6x - 2 = 4x + 8$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

Identify the independent and dependent variables

MAIN Grade 6 main question

6.EE.C.9

BACKWARD 1 Input-output rules

4.OA.C.5

BACKWARD 2 Two related number patterns

5.OA.B.3

FORWARD 1 Write an equation for a proportional relationship

7.RP.A.2.c

FORWARD 2 Functions: each input has exactly one output

8.F.A.1

SET 56 • 6.EE.C.9

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.C.9

The equation $c = 2.5p$ gives the cost c of p pounds of grapes.

Which is the independent variable?

- A.** p (pounds)
- B.** c (cost)
- C.** 2.5
- D.** There is none.

Money earned m depends on the number of hours worked h .

Which is the dependent variable?

Dependent variable: _____

A plant's height depends on the number of days it grows.

Which is the independent variable?

- A.** The plant's height
- B.** The number of days
- C.** Neither
- D.** Both

In $y = 4x + 3$, which variable is the dependent variable?

Dependent variable: _____

True or false?

Ice cream sales s depend on the temperature t .
So t is the independent variable.

- A.** True
- B.** False

SET 56 • 6.EE.C.9

BACKWARD 1

Input-output rules • Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5

The rule is "multiply by 3."

What is the output when the input is 7?

Output: _____

True or false?

The rule is "add 6." When the input is 4, the output is 10.

A. True

B. False

What is the rule?

In	2	4	6
Out	6	8	10

A. Multiply by 2

B. Multiply by 3

C. Add 2

D. Add 4

The rule is "subtract 2."

What is the output when the input is 15?

Output: _____

True or false?

In an input-output table, the output depends on the input.

- A.** True
- B.** False

SET 56 • 6.EE.C.9

BACKWARD 2

Two related number patterns • Grade 5 • Operations & Algebraic Thinking •
5.OA.B.3

True or false?

Pattern A: 0, 2, 4, 6

Pattern B: 0, 6, 12, 18

Each number in Pattern B is 3 times the matching number in Pattern A.

A. True

B. False

The rule is "add 5," starting at 0.

What number comes next? 0, 5, 10, ____

Number: _____

Pattern A starts at 0 and adds 3. Pattern B starts at 0 and adds 9.

When Pattern A is 12, what is the matching number in Pattern B?

Number: _____

Pattern A starts at 0 and adds 1. Pattern B starts at 0 and adds 4.

Which ordered pairs come from the patterns?

A. (0, 4), (1, 5), (2, 6)

B. (0, 0), (1, 4), (2, 8)

C. (1, 4), (2, 5), (3, 6)

D. (0, 0), (4, 1), (8, 2)

True or false?

Pattern A: 0, 1, 2, 3

Pattern B: 0, 5, 10, 15

Each number in Pattern B is 5 more than the matching number in Pattern A.

A. True

B. False

SET 56 • 6.EE.C.9

FORWARD 1

Write an equation for a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c

Tomatoes cost \$3.20 per pound.

Write an equation for the cost c of p pounds.

Equation: _____

The table shows a proportional relationship.
Write an equation for the cost c of n ounces.

Ounces (n)	4	10
Cost (c)	\$1.20	\$3.00

Equation: _____

Which equation represents a proportional relationship?

A. $y = 4.5x$

B. $y = 4.5x + 1$

C. $y = x + 4.5$

D. $y = 4.5$

A car uses 3 gallons of gas to travel 84 miles.

Write an equation for the miles m traveled on g gallons.

Equation: _____

True or false?

$y = \frac{2}{3}x$ represents a proportional relationship.

A. True

B. False

SET 56 • 6.EE.C.9

FORWARD 2

Functions: each input has exactly one output • Grade 8 • Functions • 8.F.A.1

True or false?

A function assigns exactly one output to each input.

- A.** True
- B.** False

Which table does NOT represent a function?

A. x: 0, 1, 2 y: 0, 2, 4

B. x: 1, 1, 2 y: 3, 5, 7

C. x: 2, 3, 4 y: 5, 6, 7

D. x: 1, 2, 3 y: 4, 4, 4

In the function $C = 12t$, which variable is the input?

Input: _____

True or false?

The set of ordered pairs $(1, 2)$, $(2, 3)$, $(1, 4)$ represents a function.

- A. True
- B. False

A function gives the height of a ball after t seconds.
What is the input?

- A.** The speed
- B.** The ball
- C.** The height
- D.** The time t

Write an equation for the relationship shown in a table or graph

MAIN	Grade 6 main question	6.EE.C.9
BACKWARD 1	Finding a rule from a table	4.OA.C.5
BACKWARD 2	Reading points on a graph	5.G.A.2
BACKWARD 3	Ordered pairs from two patterns	5.OA.B.3
FORWARD 1	Equations for proportional relationships from tables and graphs	7.RP.A.2.c
FORWARD 2	Write a linear function from a table or graph	8.F.B.4

SET 57 • 6.EE.C.9

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.C.9

Write an equation that shows how y depends on x .

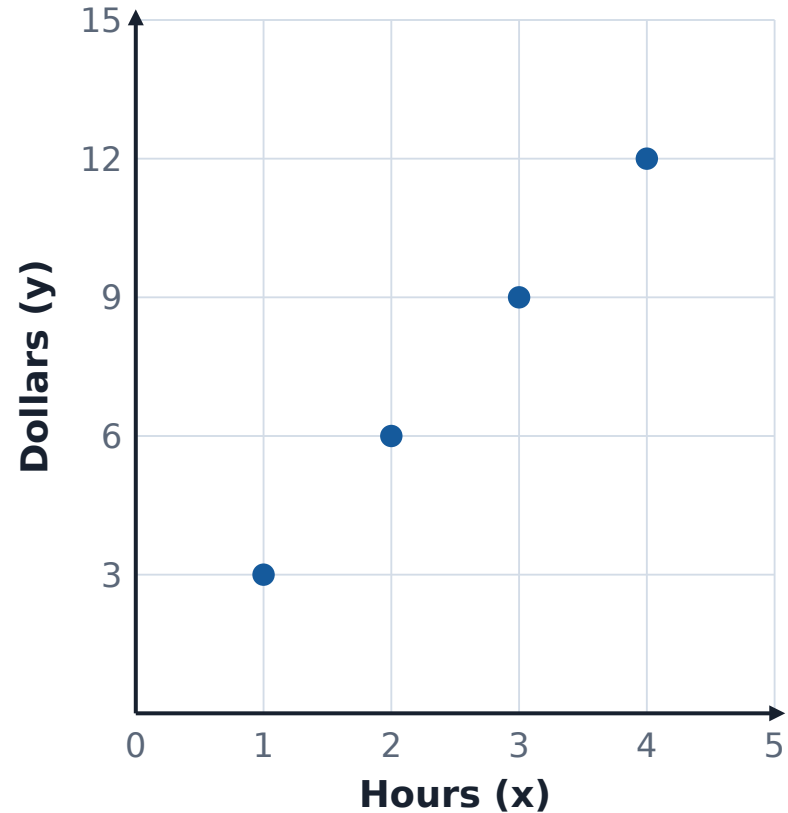
x	1	2	3	4
y	6	12	18	24

$y =$ _____

The graph shows y dollars earned for x hours worked.

Write an equation for the relationship.

$y =$ _____



The table shows a plant's height h after w weeks.
Write an equation for h in terms of w .

Weeks (w)	0	1	2	3
Height (h)	5	7	9	11

$h =$ _____

Which equation matches the table?

x	1	2	3
y	4	5	6

A. $y = 4x$

B. $y = x + 4$

C. $y = x + 3$

D. $y = 3x$

The table shows the cost c of n bags of popcorn.
Write an equation for c in terms of n .

Bags (n)	2	4	6
Cost (c)	7	14	21

$C =$ _____

SET 57 • 6.EE.C.9

BACKWARD 1

Finding a rule from a table • Grade 4 • Operations & Algebraic Thinking •
4.OA.C.5

What is the rule?

In	1	2	3	4
Out	5	10	15	20

Rule: _____

What is the rule?

In	3	5	7
Out	7	9	11

Rule: _____

True or false?

The rule for the table is "multiply by 4."

In	1	2	3
Out	4	8	12

A. True

B. False

What is the rule?

In	3	6	9
Out	1	2	3

A. Multiply by 3

B. Subtract 2

C. Add 3

D. Divide by 3

The rule is "multiply by 6." What is the output when the input is 5?

Output: _____

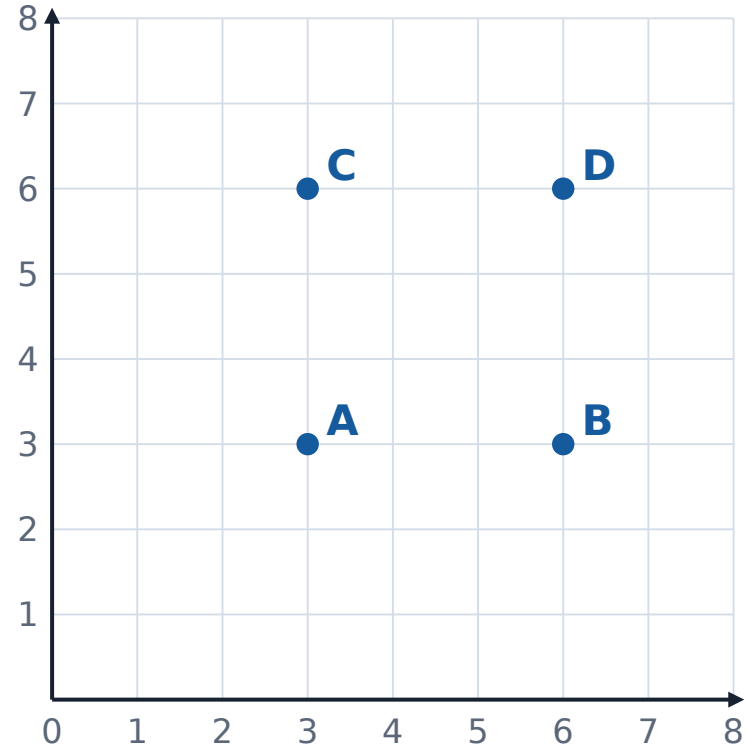
SET 57 • 6.EE.C.9

BACKWARD 2

Reading points on a graph • Grade 5 • Geometry • 5.G.A.2

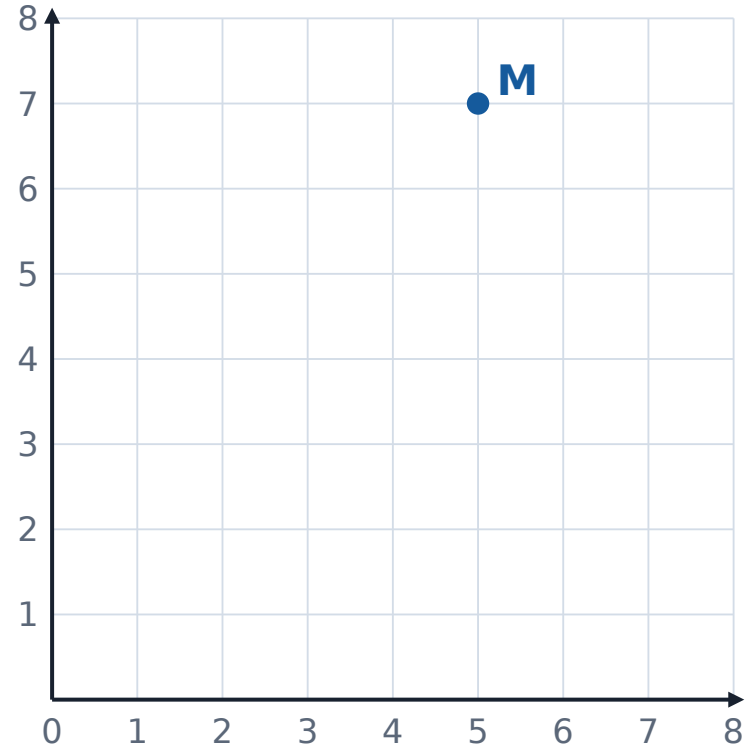
Which point is at (3, 6)?

- A.** Point A
- B.** Point B
- C.** Point C
- D.** Point D



What are the coordinates of point M?

M = _____



True or false?

The point (2, 8) is 2 units right and 8 units up from the origin.

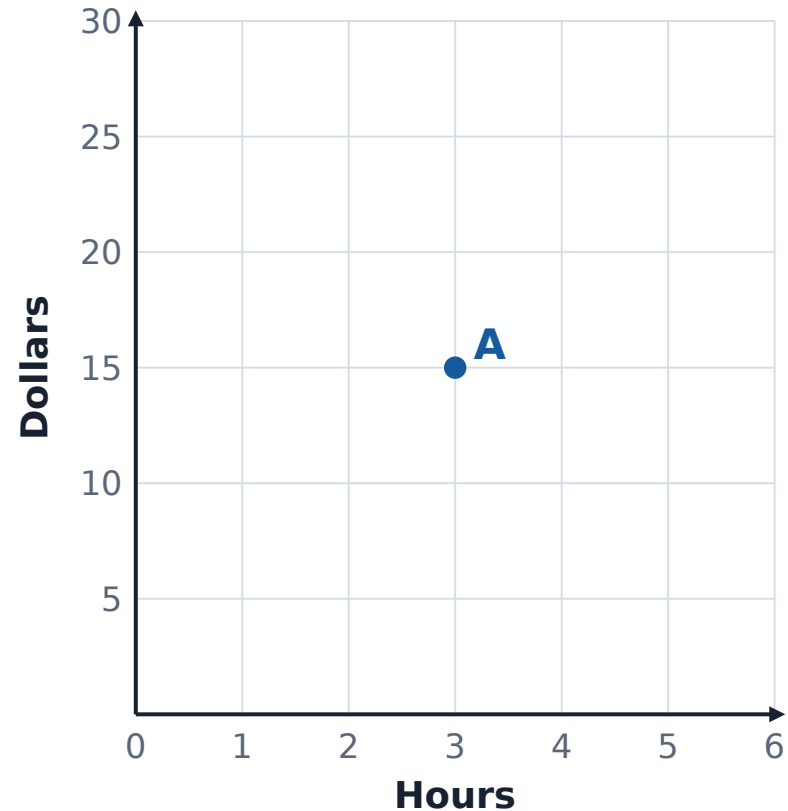
A. True

B. False

The graph shows dollars earned after working some hours.

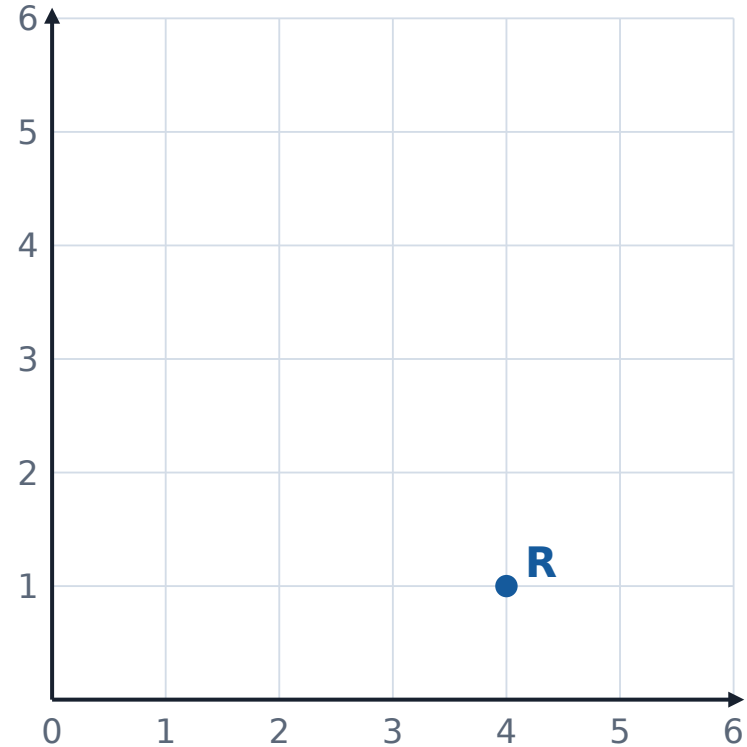
How many dollars were earned after 3 hours?

Dollars: _____



Which ordered pair names point R?

- A.** (4, 4)
- B.** (1, 4)
- C.** (1, 1)
- D.** (4, 1)



SET 57 • 6.EE.C.9

BACKWARD 3

Ordered pairs from two patterns • Grade 5 • Operations & Algebraic Thinking •
5.OA.B.3

Pattern A: 0, 1, 2, 3

Pattern B: 0, 5, 10, 15

Write the ordered pair formed by the 4th numbers.

Ordered pair: _____

True or false?

The pairs (1, 3), (2, 6), (3, 9) show that each y-value is 3 times the x-value.

A. True

B. False

Pattern A adds 2. Pattern B adds 6. Both start at 0.

How does each B number compare to its A number?

- A.** 2 more
- B.** 3 times as large
- C.** 4 more
- D.** 6 times as large

Pattern A: 0, 2, 4

Pattern B: 0, 14, 28

What is the rule that changes each A number into its B number?

Rule: _____

True or false?

The pairs $(0, 0)$, $(1, 4)$, $(2, 8)$ come from Pattern A "add 4" and Pattern B "add 1."

- A. True
- B. False

SET 57 • 6.EE.C.9

FORWARD 1

Equations for proportional relationships from tables and graphs • Grade 7 •
Ratios & Proportional Relationships • 7.RP.A.2.c

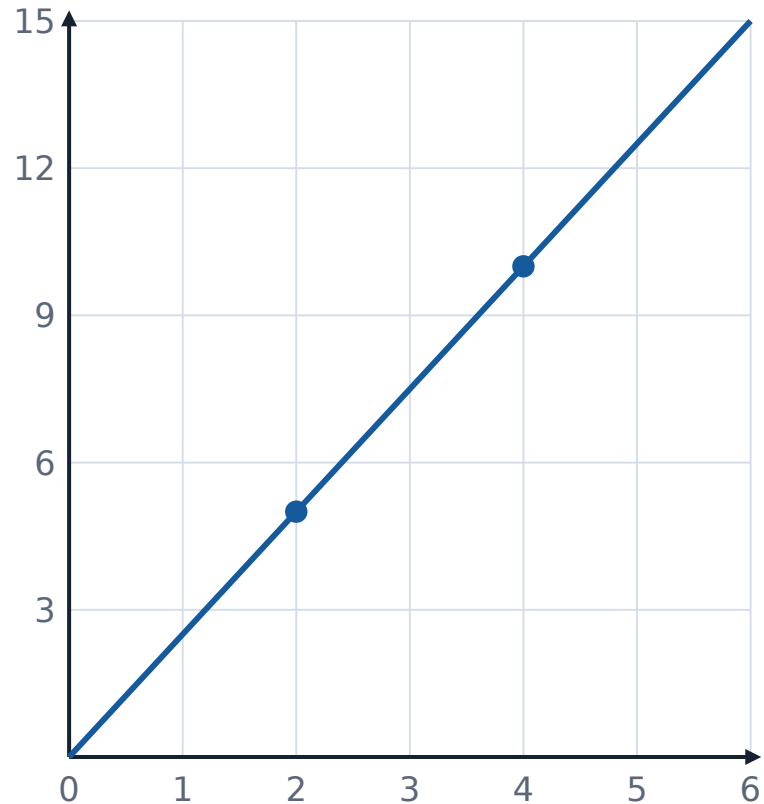
Write an equation for the proportional relationship in the table.

x	2	5	8
y	7	17.5	28

$y =$ _____

Write an equation for the proportional relationship in the graph.

$y =$ _____



The table shows a proportional relationship.
Which equation matches?

x	4	8	12
y	3	6	9

A. $y = 0.75x$

B. $y = 1.33x$

C. $y = 3x$

D. $y = x - 1$

Write an equation for the cost c of p pounds.

Pounds (p)	3	6	9
Cost (c)	7.50	15.00	22.50

$C =$ _____

True or false?

The table shows the equation $y = 6x$.

x	1	2	4
y	6	12	24

A. True

B. False

SET 57 • 6.EE.C.9

FORWARD 2

Write a linear function from a table or graph • Grade 8 • Functions • 8.F.B.4

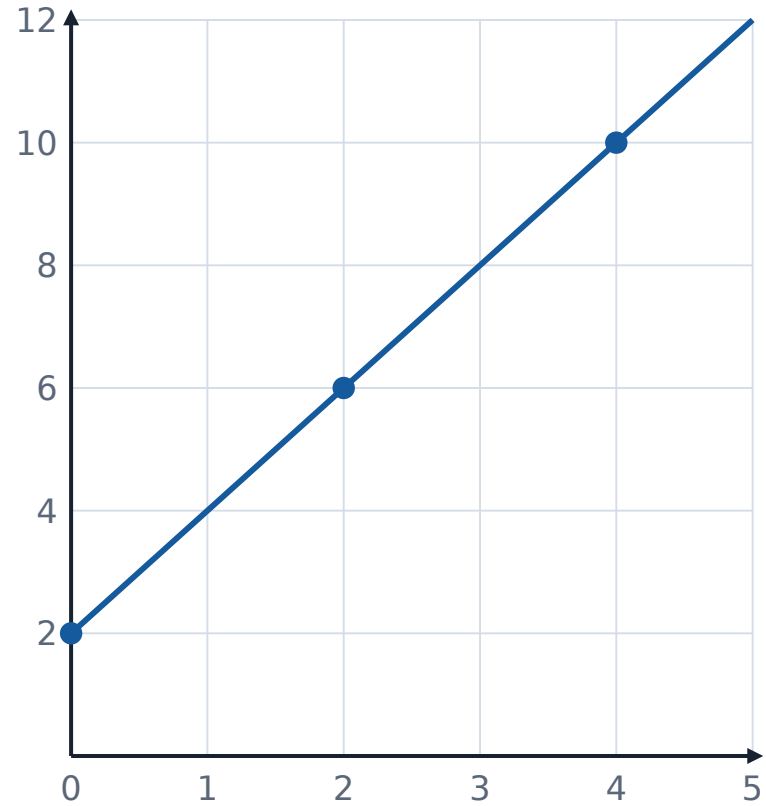
Write a function for the table.

x	0	1	2	3
y	4	7	10	13

$y =$ _____

Write a function for the line in the graph.

$y =$ _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 57 • FORWARD 2 • Q2/5

Which function matches the table?

x	0	1	2
y	10	8	6

A. $y = -2x + 10$

B. $y = 10x - 2$

C. $y = -2x$

D. $y = 2x + 10$

Write a function for the table.

x	1	2	3	4
y	9	13	17	21

$y =$ _____

True or false?

The line through $(0, 1)$ and $(3, 7)$ has the equation $y = 2x + 1$.

A. True

B. False

Area of right triangles and other triangles

MAIN

Grade 6 main question

6.G.A.1

BACKWARD 1

Area of rectangles

3.MD.C.7.b

BACKWARD 2

Half of a product

5.NF.B.4.a

BACKWARD 3

Right angles and perpendicular segments

4.G.A.1

FORWARD 1

Area and volume of figures made of triangles

7.G.B.6

FORWARD 2

Use the Pythagorean Theorem in right triangles

8.G.B.7

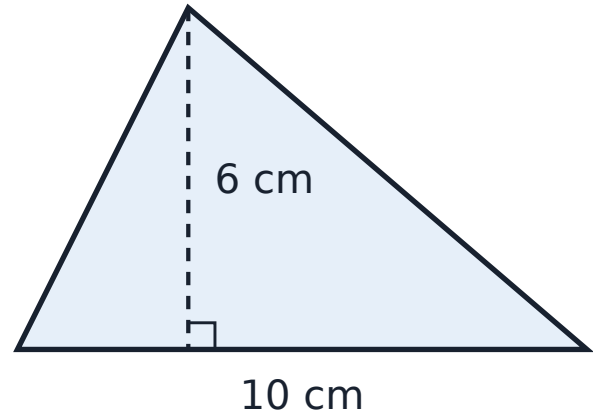
SET 58 • 6.G.A.1

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.1

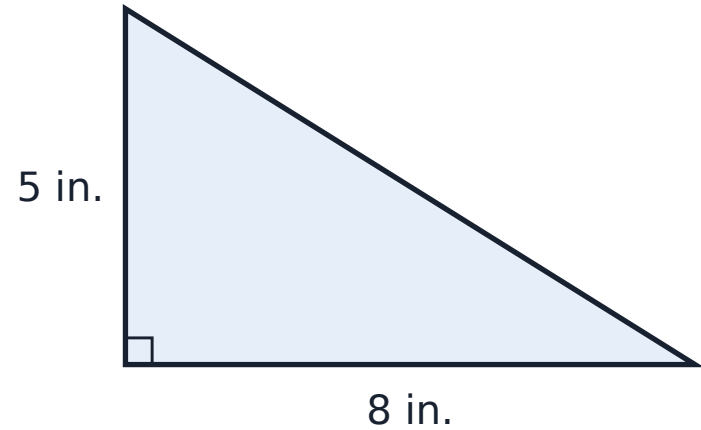
Find the area of the triangle.

Area: _____



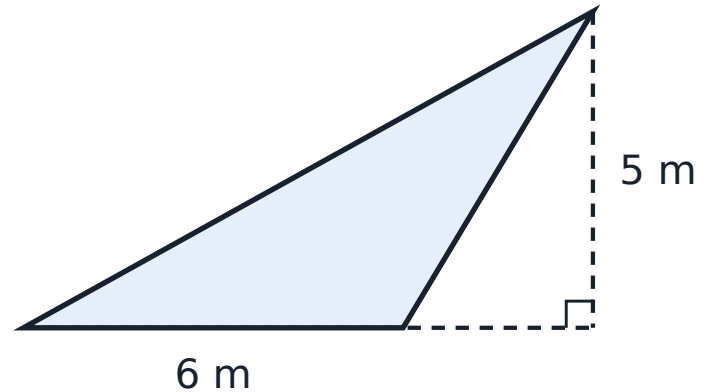
Find the area of the right triangle.

Area: _____



Find the area of the triangle.

Area: _____



A triangle has a base of 12 feet and a height of 7 feet.

What is its area?

- A.** 19 square feet
- B.** 38 square feet
- C.** 42 square feet
- D.** 84 square feet

A triangular sign has a base of 3.5 feet and a height of 2 feet.

What is its area?

Area: _____

SET 58 • 6.G.A.1

BACKWARD 1

Area of rectangles • Grade 3 • Measurement & Data • 3.MD.C.7.b

Find the area of the rectangle.

Area: _____



A rectangle is 9 feet long and 6 feet wide.
What is its area?

Area: _____

True or false?

A rectangle that is 5 cm long and 3 cm wide has an area of 15 square centimeters.

- A. True
- B. False

A rectangle is 12 m long and 4 m wide.

What is its area?

- A.** 16 square meters
- B.** 24 square meters
- C.** 32 square meters
- D.** 48 square meters

Find the area of the rectangle.

Area: _____



SET 58 • 6.G.A.1

BACKWARD 2

Half of a product • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$$\frac{1}{2} \times 10 \times 6$$

Product: _____

Multiply.

$$\frac{1}{2} \times 14$$

Product: _____

True or false?

$$\frac{1}{2} \times 8 \times 5 = 20$$

A. True

B. False

$$\frac{1}{2} \times 12 \times 7 = ?$$

A. 19

B. 21

C. 42

D. 84

Multiply.

$$\frac{1}{2} \times 9 \times 4$$

Product: _____

SET 58 • 6.G.A.1

BACKWARD 3

Right angles and perpendicular segments • Grade 4 • Geometry • 4.G.A.1

True or false?

Perpendicular lines meet to form right angles.

A. True

B. False

Which angle measure is a right angle?

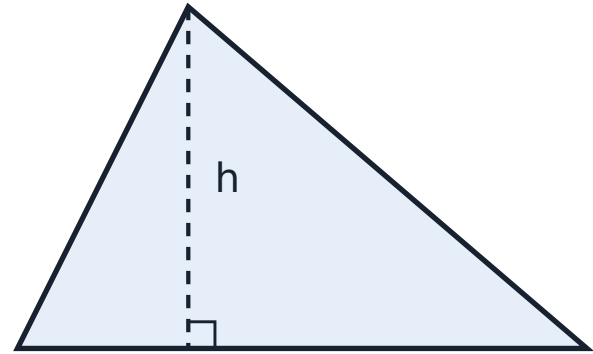
- A.** 45°
- B.** 90°
- C.** 100°
- D.** 180°

True or false?

The dashed segment is perpendicular to the base of the triangle.

A. True

B. False



How many degrees are in a right angle?

Degrees: _____

True or false?

A square has 4 right angles.

A. True

B. False

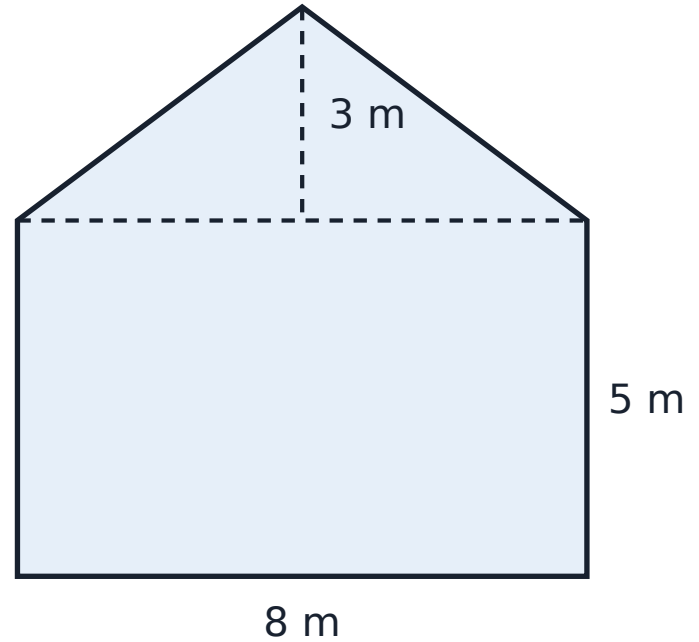
SET 58 • 6.G.A.1

FORWARD 1

Area and volume of figures made of triangles • Grade 7 • Geometry • 7.G.B.6

Find the area of the figure.

Area: _____



A kite is made of two triangles. Each triangle has a base of 10 inches and a height of 4 inches.

What is the area of the kite?

Area: _____

A triangle with a base of 4 cm and a height of 3 cm is cut out of a rectangle that is 12 cm by 6 cm.

What is the area of the part that is left?

- A.** 60 square cm
- B.** 66 square cm
- C.** 72 square cm
- D.** 78 square cm

A triangular prism has triangle bases with a base of 6 cm and a height of 4 cm. The prism is 10 cm long.

What is its volume?

Volume: _____

True or false?

A figure made of a 6-by-4 rectangle and a triangle with a base of 3 and a height of 4 has an area of 30 square units.

- A.** True
- B.** False

SET 58 • 6.G.A.1

FORWARD 2

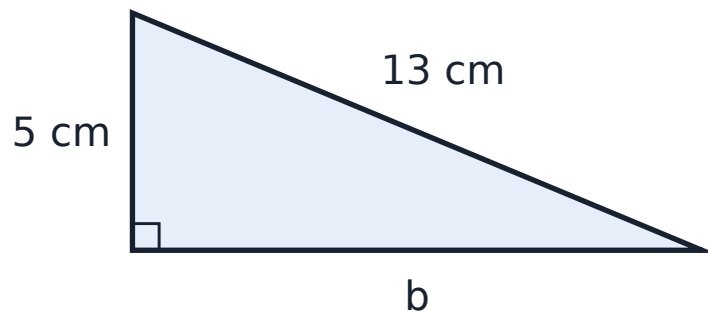
Use the Pythagorean Theorem in right triangles • Grade 8 • Geometry • 8.G.B.7

A right triangle has legs of 6 cm and 8 cm.
What is the length of the hypotenuse?

Hypotenuse: _____

Find the length b .

$b =$ _____



A 10-foot ladder leans against a wall. The top reaches 8 feet up the wall.

How far is the bottom of the ladder from the wall?

A. 2 feet

B. 6 feet

C. 12.8 feet

D. 18 feet

An isosceles triangle has a base of 16 inches and two sides of 10 inches.

What is its area?

Area: _____

A right triangle has legs of 9 m and 12 m.
What is the length of the hypotenuse?

Hypotenuse: _____

Area of parallelograms and trapezoids

MAIN	Grade 6 main question	6.G.A.1
BACKWARD 1	Area of rectangles	3.MD.C.7.b
BACKWARD 2	Area by splitting into rectangles	3.MD.C.7.d
BACKWARD 3	Classifying quadrilaterals	5.G.B.4
BACKWARD 4	Height is perpendicular to the base	4.G.A.1
FORWARD 1	Area of figures made of quadrilaterals in real-world problems	7.G.B.6
FORWARD 2	Use the Pythagorean Theorem to find a missing height	8.G.B.7

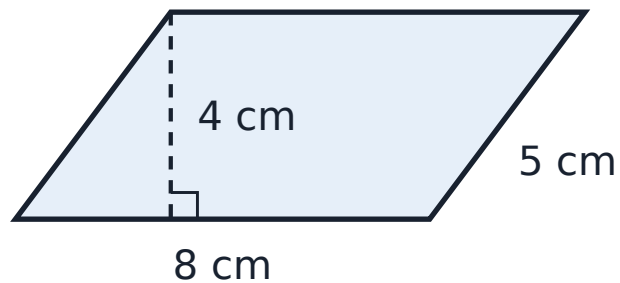
SET 59 • 6.G.A.1

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.1

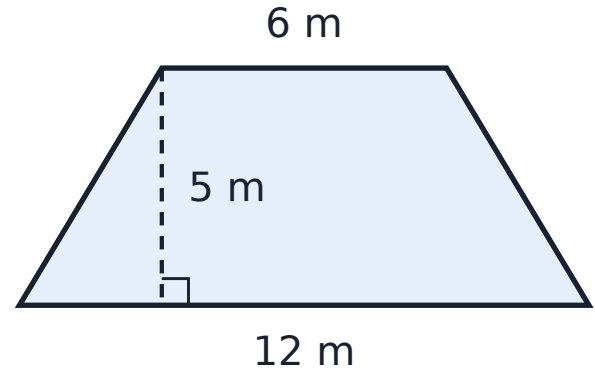
Find the area of the
parallelogram.

Area: _____



Find the area of the trapezoid.

Area: _____



A parallelogram has a base of 9 feet and a height of 5 feet.

What is its area?

A. 14 square feet

B. 22.5 square feet

C. 28 square feet

D. 45 square feet

A trapezoid has bases of 7 inches and 11 inches. Its height is 4 inches.

What is its area?

Area: _____

A parallelogram has a base of 12.5 meters and a height of 4 meters.

What is its area?

Area: _____

SET 59 • 6.G.A.1

BACKWARD 1

Area of rectangles • Grade 3 • Measurement & Data • 3.MD.C.7.b

Find the area of the rectangle.

Area: _____



A rectangle is 11 cm long and 5 cm wide.

What is its area?

Area: _____

True or false?

A square with 6-meter sides has an area of 24 square meters.

A. True

B. False

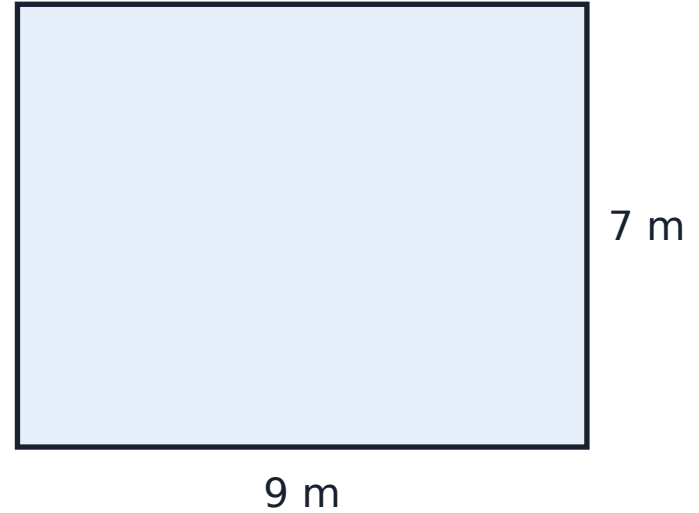
A rectangle is 15 inches long and 2 inches wide.

What is its area?

- A.** 17 square inches
- B.** 30 square inches
- C.** 34 square inches
- D.** 60 square inches

Find the area of the rectangle.

Area: _____



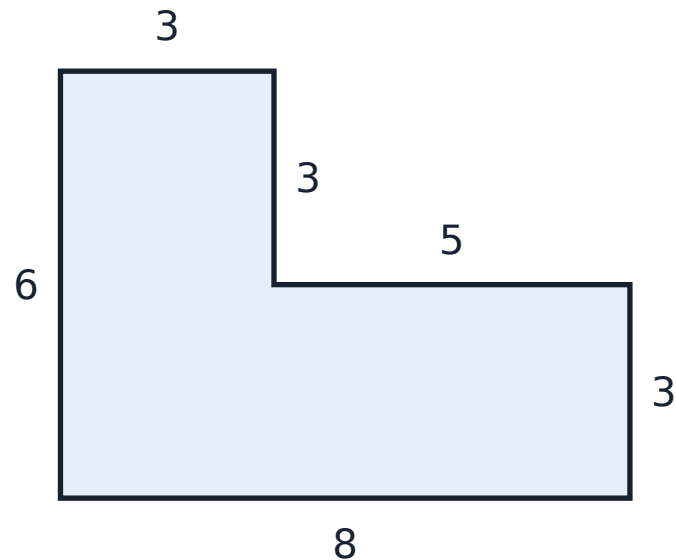
SET 59 • 6.G.A.1

BACKWARD 2

Area by splitting into rectangles • Grade 3 • Measurement & Data • 3.MD.C.7.d

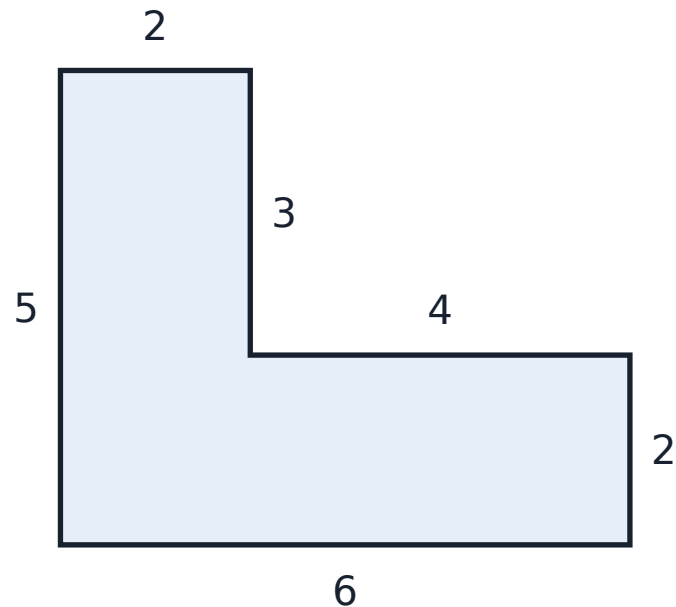
Find the area of the figure.

Area: _____



Find the area of the figure.

Area: _____



True or false?

The area of an L-shaped figure can be found by adding the areas of two rectangles.

- A.** True
- B.** False

A figure is made of a 4-by-5 rectangle and a 2-by-3 rectangle that do not overlap.

What is its area?

A. 14 square units

B. 20 square units

C. 26 square units

D. 120 square units

A figure is made of a 10-by-2 rectangle and a 3-by-3 square that do not overlap.

What is its area?

Area: _____

SET 59 • 6.G.A.1

BACKWARD 3

Classifying quadrilaterals • Grade 5 • Geometry • 5.G.B.4

True or false?

Every rectangle is a parallelogram.

A. True

B. False

True or false?

Every parallelogram is a rectangle.

A. True

B. False

Which quadrilateral always has four right angles?

- A.** Trapezoid
- B.** Rhombus
- C.** Parallelogram
- D.** Rectangle

How many pairs of parallel sides does a parallelogram have?

A. 0

B. 1

C. 2

D. 4

True or false?

A square is a rhombus.

A. True

B. False

SET 59 • 6.G.A.1

BACKWARD 4

Height is perpendicular to the base • Grade 4 • Geometry • 4.G.A.1

True or false?

The height of a parallelogram is perpendicular to its base.

A. True

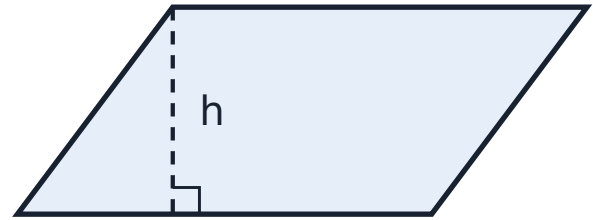
B. False

True or false?

The dashed segment shows the height of the parallelogram.

A. True

B. False



Which statement describes perpendicular lines?

- A.** They meet at right angles.
- B.** They never meet.
- C.** They meet at a 45° angle.
- D.** They are always the same length.

True or false?

The slanted side of a parallelogram is always its height.

A. True

B. False

What is the measure of the angle between a base and its height?

Angle: _____

SET 59 • 6.G.A.1

FORWARD 1

Area of figures made of quadrilaterals in real-world problems • Grade 7 •
Geometry • 7.G.B.6

A patio is made of a rectangle that is 10 ft by 6 ft and a trapezoid with bases of 6 ft and 4 ft and a height of 3 ft.
What is the area of the patio?

Area: _____

A wall is 12 ft by 8 ft. It has a window shaped like a parallelogram with a base of 3 ft and a height of 2 ft.

What is the area of the wall that is not window?

Area: _____

A sign is a trapezoid with bases of 1.5 m and 2.5 m and a height of 1.2 m.

What is its area?

- A.** 2.4 square meters
- B.** 3.6 square meters
- C.** 4.8 square meters
- D.** 5.2 square meters

A garden is made of two parallelograms that do not overlap. Each has a base of 7 m and a height of 4 m.

What is the total area?

Area: _____

True or false?

A trapezoid with bases of 6 and 10 and a height of 5 has the same area as a rectangle that is 8 by 5.

- A. True
- B. False

SET 59 • 6.G.A.1

FORWARD 2

Use the Pythagorean Theorem to find a missing height • Grade 8 • Geometry •
8.G.B.7

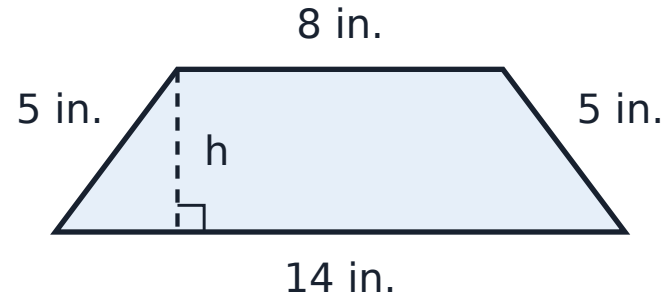
A parallelogram has a slanted side of 5 cm. The slanted side, the height, and a 3 cm piece of the base form a right triangle.

What is the height?

Height: _____

Find the height h of the trapezoid.

$h =$ _____



A ramp forms a right triangle. The ramp is 13 ft long and its base is 12 ft long.

How tall is the ramp?

A. 1 ft

B. 5 ft

C. 17.7 ft

D. 25 ft

A parallelogram has a slanted side of 10 m. The slanted side, the height, and a 6 m piece of the base form a right triangle.

What is the height?

Height: _____

True or false?

A trapezoid has a slanted side of 10 in. The slanted side, the height, and a 6 in. piece of the base form a right triangle. The height is 8 in.

A. True

B. False

Area of polygons by composing and decomposing

MAIN	Grade 6 main question	6.G.A.1
BACKWARD 1	Area of rectilinear figures	3.MD.C.7.d
BACKWARD 2	Missing side lengths of rectangles	4.MD.A.3
BACKWARD 3	Area of triangles	6.G.A.1
FORWARD 1	Area of composite figures in real-world problems	7.G.B.6
FORWARD 2	Volume of composite solids	8.G.C.9

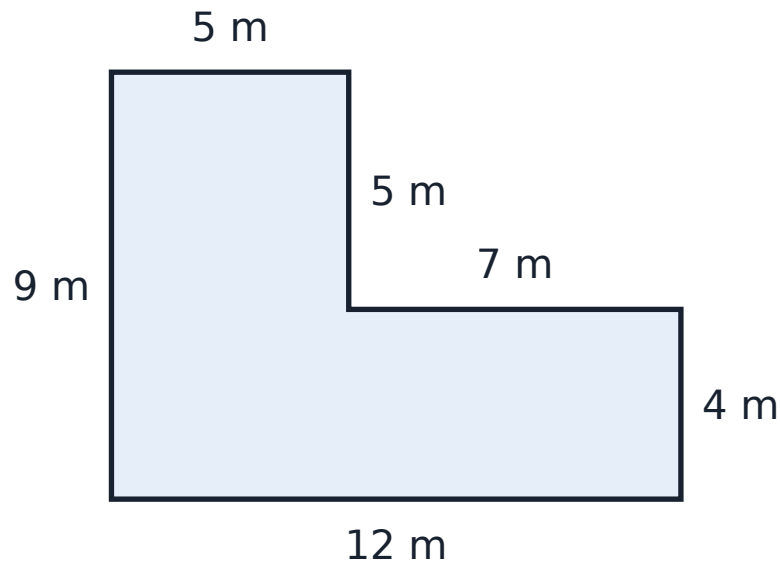
SET 60 • 6.G.A.1

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.1

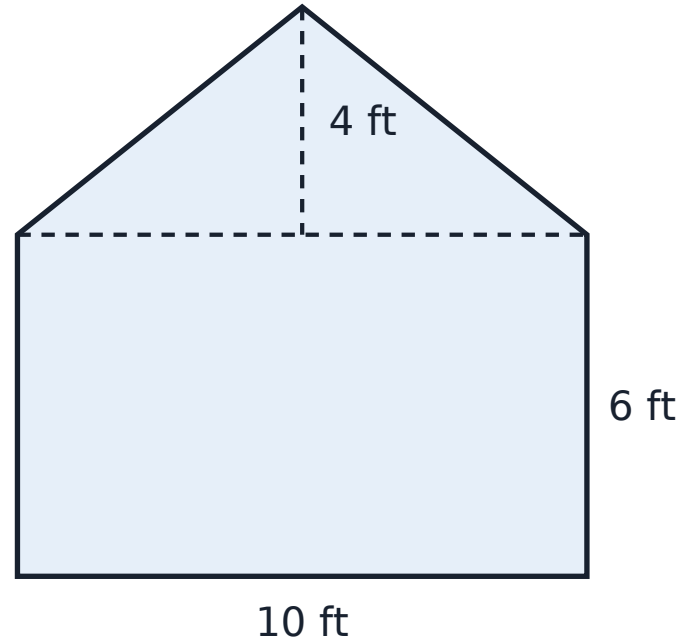
Find the area of the figure.

Area: _____



Find the area of the figure.

Area: _____



A rectangular yard is 20 m by 15 m. A triangular garden in the yard has a base of 6 m and a height of 5 m.

What is the area of the yard that is NOT garden?

Area: _____

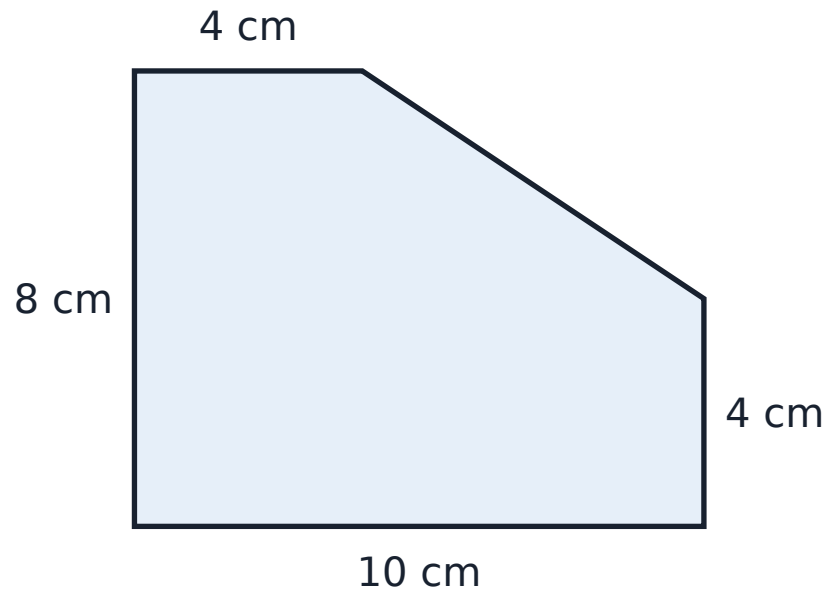
A polygon is split into a rectangle that is 8 by 3 and two triangles. Each triangle has a base of 2 and a height of 3.

What is the area of the polygon?

- A.** 24 square units
- B.** 27 square units
- C.** 30 square units
- D.** 36 square units

Find the area of the figure.

Area: _____



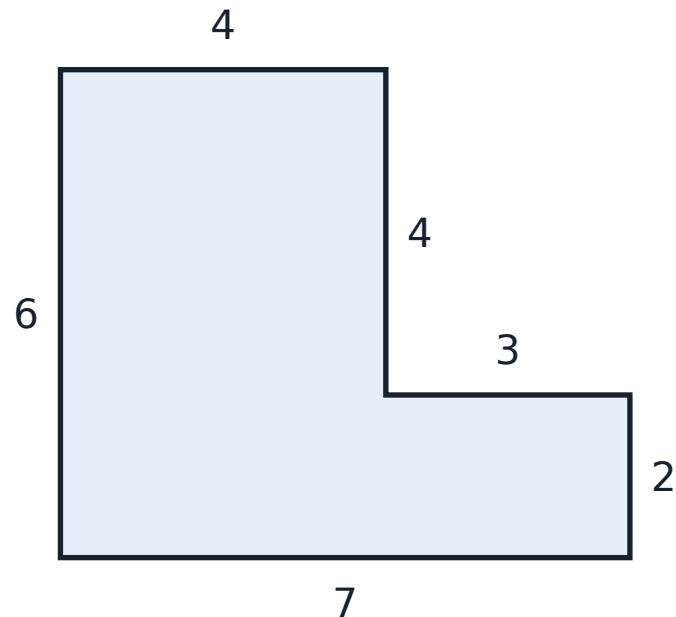
SET 60 • 6.G.A.1

BACKWARD 1

Area of rectilinear figures • Grade 3 • Measurement & Data • 3.MD.C.7.d

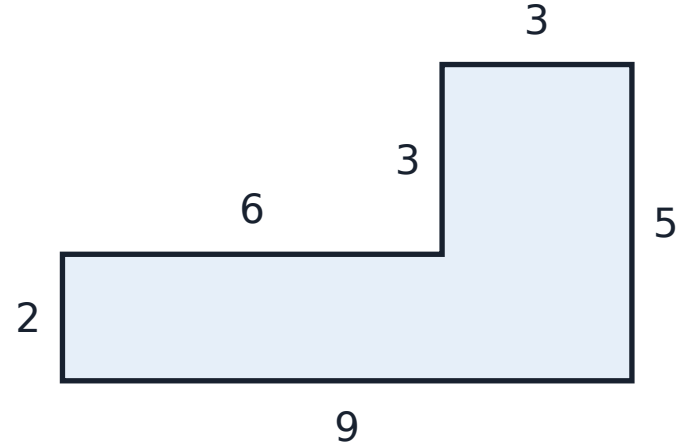
Find the area of the figure.

Area: _____



Find the area of the figure.

Area: _____



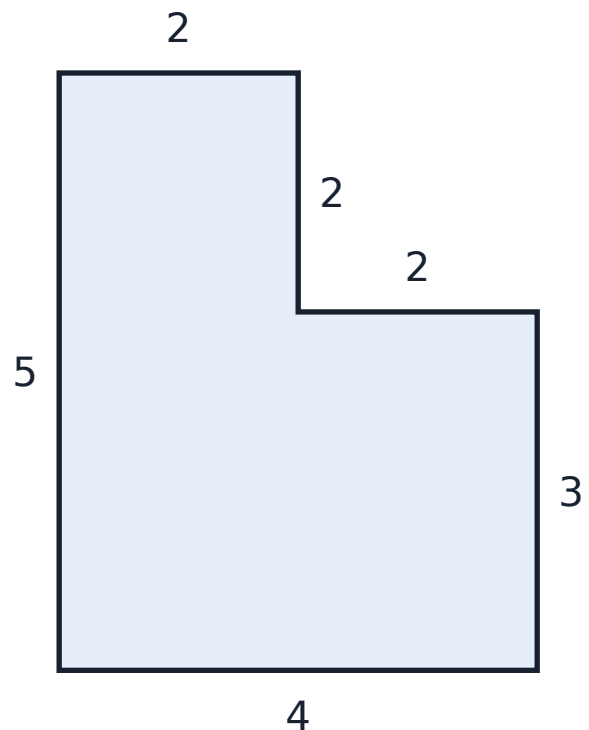
True or false?

A figure made of a 5-by-2 rectangle and a 3-by-4 rectangle has an area of 22 square units.

- A. True
- B. False

Which expression gives the area of the figure?

- A.** 4×5
- B.** $4 + 3 + 2 + 2$
- C.** $4 \times 3 \times 2$
- D.** $4 \times 3 + 2 \times 2$



A figure is made of two rectangles: 6 by 3 and 2 by 4.
What is its area?

Area: _____

SET 60 • 6.G.A.1

BACKWARD 2

Missing side lengths of rectangles • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle has an area of 48 square feet and a width of 6 feet.

What is its length?

Length: _____

A rectangle has a perimeter of 30 cm and a length of 9 cm.

What is its width?

Width: _____

True or false?

A rectangle with an area of 36 square cm and a length of 9 cm has a width of 4 cm.

A. True

B. False

A rectangle has an area of 56 square inches and a width of 7 inches.

What is its length?

A. 8 inches

B. 9 inches

C. 49 inches

D. 63 inches

The rectangle has an area of 60 square meters.

What is the missing side length?



Length: _____

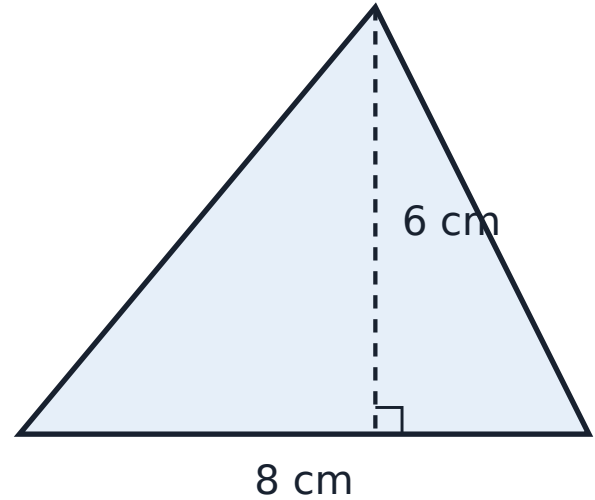
SET 60 • 6.G.A.1

BACKWARD 3

Area of triangles • Grade 6 • Geometry • 6.G.A.1

Find the area of the triangle.

Area: _____



A triangle has a base of 10 m and a height of 3 m.

What is its area?

Area: _____

True or false?

A right triangle with legs of 4 in. and 6 in. has an area of 24 square inches.

A. True

B. False

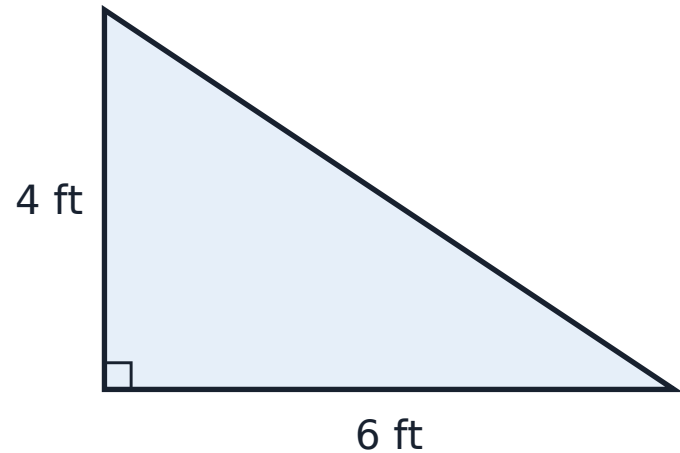
A triangle has a base of 5 ft and a height of 8 ft.

What is its area?

- A.** 13 square feet
- B.** 20 square feet
- C.** 26 square feet
- D.** 40 square feet

Find the area of the triangle.

Area: _____



SET 60 • 6.G.A.1

FORWARD 1

Area of composite figures in real-world problems • Grade 7 • Geometry •
7.G.B.6

A deck is made of a rectangle that is 12 ft by 10 ft and a triangle with a base of 10 ft and a height of 6 ft.

What is the area of the deck?

Area: _____

A trapezoid-shaped garden has bases of 8 m and 14 m and a height of 5 m. It has a square pond with 2 m sides.

What is the area of the garden without the pond?

Area: _____

A regular hexagon is split into 6 equal triangles. Each triangle has a base of 4 cm and a height of 3.5 cm.

What is the area of the hexagon?

- A.** 21 square cm
- B.** 24 square cm
- C.** 42 square cm
- D.** 84 square cm

An L-shaped prism has a base area of 30 square feet and a height of 4 feet.

What is its volume?

Volume: _____

True or false?

A figure made of a 5-by-5 square and a triangle with a base of 5 and a height of 2 has an area of 30 square units.

- A.** True
- B.** False

SET 60 • 6.G.A.1

FORWARD 2

Volume of composite solids • Grade 8 • Geometry • 8.G.C.9

A silo is a cylinder with a radius of 3 m and a height of 10 m, topped by a hemisphere with a radius of 3 m.

What is its volume in terms of π ?

Volume: _____

A cone has a radius of 3 in. and a height of 4 in. A hemisphere with a radius of 3 in. sits on top.

What is the total volume in terms of π ?

Volume: _____

A cone with a radius of 2 and a height of 3 is cut out of a cylinder with a radius of 2 and a height of 6.

What volume is left?

A. 20π

B. 28π

C. 24π

D. 4π

A cylinder and a cone each have a radius of 1 and a height of 2.

How much more volume does the cylinder have, in terms of π ?

Difference: _____

True or false?

A sphere with a radius of 3 has the same volume as a cylinder with a radius of 3 and a height of 4.

- A. True
- B. False

Find volume by packing a prism with unit-fraction cubes

MAIN	Grade 6 main question	6.G.A.2
BACKWARD 1	Volume by counting unit cubes	5.MD.C.4
BACKWARD 2	The volume of one small cube	5.NF.B.4.a
BACKWARD 3	Volume as the number of unit cubes	5.MD.C.3.b
FORWARD 1	Volume of prisms in real-world problems	7.G.B.6
FORWARD 2	Volume of cylinders, cones, and spheres	8.G.C.9

SET 61 • 6.G.A.2

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.2

How many cubes with $\frac{1}{2}$ -inch edges fill a box that is 2 in. by $1\frac{1}{2}$ in. by 1 in.?

Cubes: _____

A box is 3 in. by 2 in. by $1\frac{1}{2}$ in.

How many cubes with $\frac{1}{2}$ -inch edges fill the box?

Cubes: _____

True or false?

A prism is packed with 48 cubes that each have $\frac{1}{2}$ -unit edges. Its volume is 6 cubic units.

- A. True
- B. False

A prism is packed with 64 cubes that each have $\frac{1}{4}$ -inch edges.

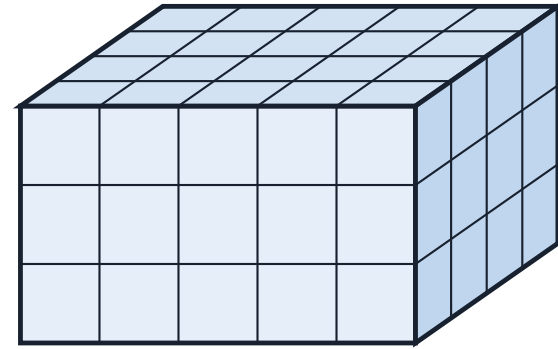
What is the volume of the prism?

- A.** $\frac{1}{4}$ cubic inch
- B.** 1 cubic inch
- C.** 4 cubic inches
- D.** 16 cubic inches

The prism is packed with cubes that have $\frac{1}{2}$ -cm edges.

It is 5 cubes long, 4 cubes wide, and 3 cubes tall.

What is its volume?



Volume: _____

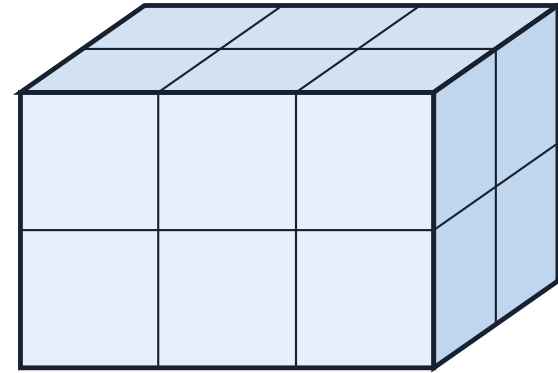
SET 61 • 6.G.A.2

BACKWARD 1

Volume by counting unit cubes • Grade 5 • Measurement & Data • 5.MD.C.4

Each cube is 1 cubic unit.

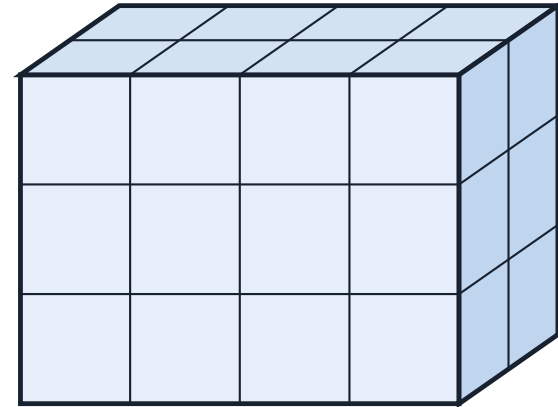
What is the volume of the prism?



Volume: _____

Each cube is 1 cubic unit.

What is the volume of the prism?



Volume: _____

True or false?

A prism made of 2 layers with 6 unit cubes in each layer has a volume of 12 cubic units.

- A.** True
- B.** False

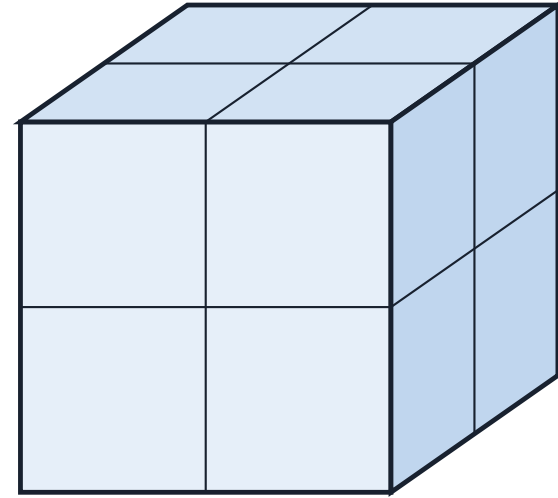
A box holds 3 layers with 5 unit cubes in each layer.

What is its volume?

- A.** 8 cubic units
- B.** 15 cubic units
- C.** 35 cubic units
- D.** 53 cubic units

Each cube is 1 cubic unit.

What is the volume of the prism?



Volume: _____

SET 61 • 6.G.A.2

BACKWARD 2

The volume of one small cube • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.a

Multiply.

$$\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$$

Product: _____

Multiply.

$$\frac{1}{4} \times \frac{1}{4} \times \frac{1}{4}$$

Product: _____

True or false?

$$\frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} = \frac{1}{9}$$

A. True

B. False

Multiply.

$$24 \times \frac{1}{8}$$

A. $\frac{1}{3}$

B. 3

C. 32

D. 192

Multiply.

$$64 \times \frac{1}{64}$$

Product: _____

SET 61 • 6.G.A.2

BACKWARD 3

Volume as the number of unit cubes • Grade 5 • Measurement & Data •
5.MD.C.3.b

True or false?

A solid that can be packed with 20 unit cubes with no gaps has a volume of 20 cubic units.

- A. True
- B. False

A box is filled with 36 unit cubes with no gaps or overlaps.

What is its volume?

- A.** 6 cubic units
- B.** 12 cubic units
- C.** 36 cubic units
- D.** 36 square units

A box is filled with 45 unit cubes with no gaps.

What is its volume?

Volume: _____

True or false?

Volume is measured in square units.

A. True

B. False

Which unit could be used to measure volume?

- A.** Square centimeters
- B.** Cubic centimeters
- C.** Centimeters
- D.** Kilograms

SET 61 • 6.G.A.2

FORWARD 1

Volume of prisms in real-world problems • Grade 7 • Geometry • 7.G.B.6

A triangular prism has triangle bases with a base of 5 in. and a height of 4 in. The prism is 8 in. long.

What is its volume?

Volume: _____

A prism has a base area of $12\frac{1}{2}$ square cm and a height of 4 cm.

What is its volume?

Volume: _____

A cube has edges of 2.5 cm.

What is its volume?

- A.** 6.25 cubic cm
- B.** 7.5 cubic cm
- C.** 15 cubic cm
- D.** 15.625 cubic cm

An aquarium is 30 in. long, 12 in. wide, and 16 in. tall. It is filled with water to a height of 12 in.

What is the volume of the water?

Volume: _____

True or false?

A triangular prism with a base area of 15 square units and a length of 6 units has a volume of 90 cubic units.

A. True

B. False

SET 61 • 6.G.A.2

FORWARD 2

Volume of cylinders, cones, and spheres • Grade 8 • Geometry • 8.G.C.9

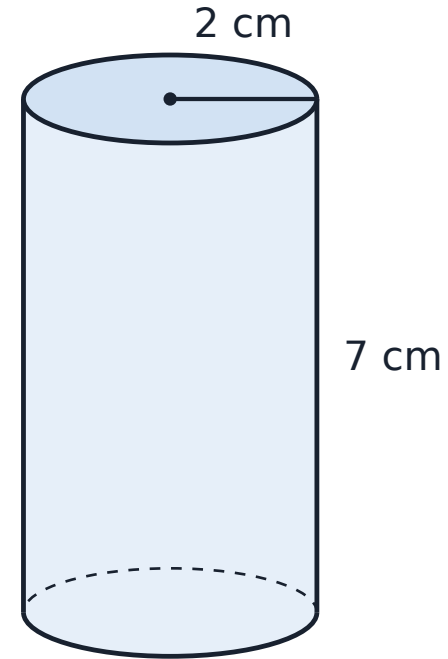
A cylinder has a radius of 5 and a height of 4.
What is its volume in terms of π ?

Volume: _____

Find the volume of the cylinder.

Use 3.14 for π .

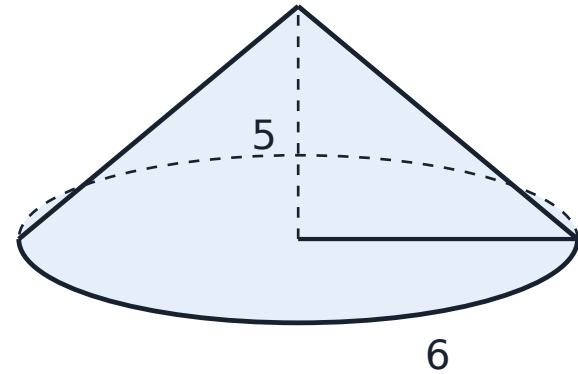
Volume: _____



A cone has a radius of 6 and a height of 5.

What is its volume?

- A.** 150π **B.** 180π
C. 30π **D.** 60π



A sphere has a radius of 6.

What is its volume in terms of π ?

Volume: _____

True or false?

A cylinder has a greater volume than a cone with the same radius and height.

A. True

B. False

Apply $V = l w h$ and $V = b h$ with fractional edge lengths

MAIN

Grade 6 main question

6.G.A.2

BACKWARD 1

Volume formula with whole numbers

5.MD.C.5.b

BACKWARD 2

Multiplying mixed numbers

5.NF.B.6

BACKWARD 3

Multiplying fractions

5.NF.B.4.a

FORWARD 1

Volume of triangular and rectangular prisms

7.G.B.6

FORWARD 2

Volume of cylinders using $V = Bh$

8.G.C.9

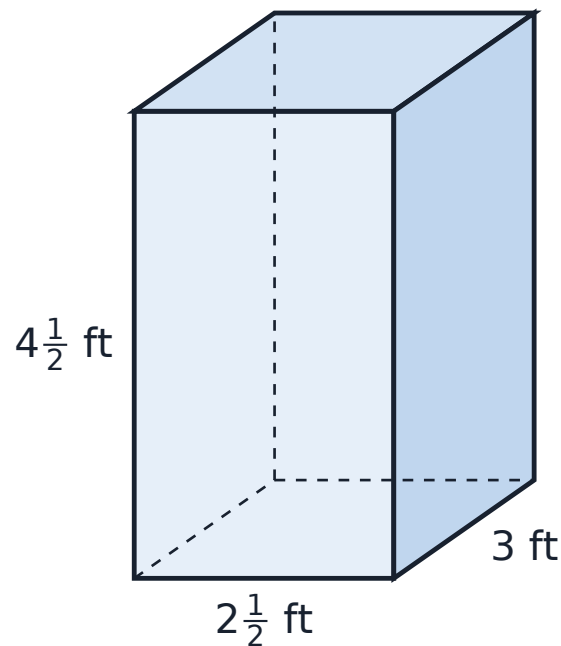
SET 62 • 6.G.A.2

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.2

Find the volume of the rectangular prism.

Volume: _____



A box is $1\frac{1}{2}$ ft long, 2 ft wide, and $\frac{3}{4}$ ft tall.

What is its volume?

- A.** $2\frac{1}{4}$ cubic feet
- B.** $2\frac{3}{4}$ cubic feet
- C.** 3 cubic feet
- D.** $4\frac{1}{4}$ cubic feet

Use $V = l \times w \times h$ to find the volume of a prism with $l = 5$ cm, $w = 2\frac{1}{2}$ cm, and $h = 1\frac{1}{5}$ cm.

$V =$ _____

A cube has edges of $1\frac{1}{2}$ feet.

What is its volume?

Volume: _____

A prism has a base area of $4\frac{1}{2}$ square inches and a height of $2\frac{2}{3}$ inches.

Use $V = b \times h$ to find its volume.

$V =$ _____

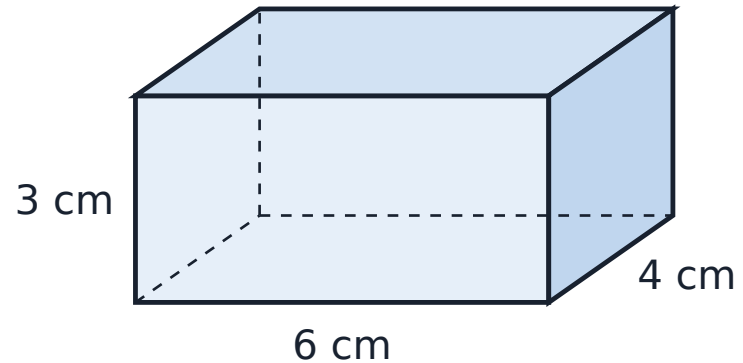
SET 62 • 6.G.A.2

BACKWARD 1

Volume formula with whole numbers • Grade 5 • Measurement & Data •
5.MD.C.5.b

Find the volume of the prism.

Volume: _____



A prism has a length of 10 m, a width of 2 m, and a height of 5 m.

What is its volume?

Volume: _____

True or false?

A cube with 3-inch edges has a volume of 27 cubic inches.

A. True

B. False

A box is 8 in. by 5 in. by 2 in.

What is its volume?

- A.** 15 cubic inches
- B.** 40 cubic inches
- C.** 80 cubic inches
- D.** 160 cubic inches

A prism has a base area of 20 square cm and a height of 6 cm.

What is its volume?

Volume: _____

SET 62 • 6.G.A.2

BACKWARD 2

Multiplying mixed numbers • Grade 5 • Number & Operations–Fractions •
5.NF.B.6

Multiply.

$$2\frac{1}{2} \times 3$$

Product: _____

Multiply.

$$1\frac{1}{2} \times \frac{3}{4}$$

Product: _____

True or false?

$$2\frac{1}{2} \times 2 = 5$$

A. True

B. False

$$3 \times 4\frac{1}{2} = ?$$

A. $7\frac{1}{2}$

B. $12\frac{1}{6}$

C. $12\frac{1}{2}$

D. $13\frac{1}{2}$

Multiply.

$$1\frac{1}{3} \times 1\frac{1}{2}$$

Product: _____

SET 62 • 6.G.A.2

BACKWARD 3

Multiplying fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$$\frac{3}{4} \times \frac{2}{3}$$

Product: _____

Multiply.

$$\frac{5}{2} \times \frac{9}{2}$$

Product: _____

True or false?

$$\frac{3}{2} \times \frac{3}{2} = \frac{9}{4}$$

A. True

B. False

$$\frac{1}{2} \times \frac{4}{5} = ?$$

A. $\frac{2}{5}$

B. $\frac{4}{7}$

C. $\frac{5}{7}$

D. $\frac{8}{5}$

Multiply.

$$\frac{6}{5} \times 5$$

Product: _____

SET 62 • 6.G.A.2

FORWARD 1

Volume of triangular and rectangular prisms • Grade 7 • Geometry • 7.G.B.6

A triangular prism has triangle bases with a base of 6 cm and a height of 4 cm. The prism is 9 cm long.

What is its volume?

Volume: _____

A tent is a triangular prism. Its triangle has a base of 2.4 m and a height of 1.5 m. The tent is 3 m long.

What is its volume?

Volume: _____

A prism has a base area of 7.5 square feet and a height of 2.4 feet.

What is its volume?

A. 9.9 cubic feet

B. 15 cubic feet

C. 18 cubic feet

D. 180 cubic feet

A box is 4.5 cm by 2 cm by 3.2 cm.

What is its volume?

Volume: _____

True or false?

A triangular prism with a triangle area of 10 square inches and a length of 7 inches has a volume of 70 cubic inches.

A. True

B. False

SET 62 • 6.G.A.2

FORWARD 2

Volume of cylinders using $V = Bh$ • Grade 8 • Geometry • 8.G.C.9

A cylinder has a base area of 12π square cm and a height of 5 cm.

What is its volume in terms of π ?

Volume: _____

A can has a radius of 3 in. and a height of 8 in.
What is its volume in terms of π ?

Volume: _____

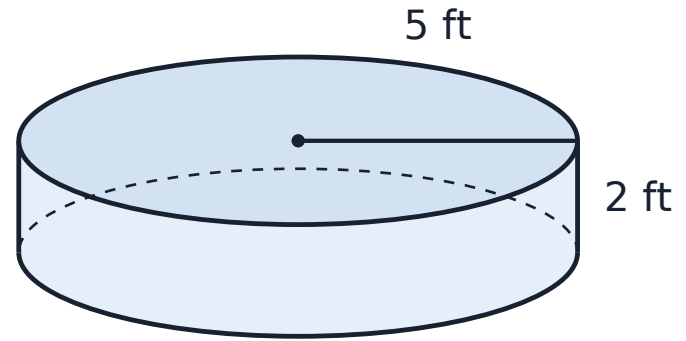
A cylinder has a radius of 1.5 m and a height of 4 m.

What is its volume?

- A.** 12π cubic meters
- B.** 9π cubic meters
- C.** 3π cubic meters
- D.** 6π cubic meters

Find the volume of the cylinder in terms of π .

Volume: _____



True or false?

Doubling the height of a cylinder doubles its volume.

A. True

B. False

Draw polygons in the coordinate plane from the coordinates of their vertices

MAIN	Grade 6 main question	6.G.A.3
BACKWARD 1	Plotting points in all four quadrants	6.NS.C.6.c
BACKWARD 2	Attributes of quadrilaterals	3.G.A.1
BACKWARD 3	Naming shapes by sides and angles	2.G.A.1
FORWARD 1	Draw shapes with given conditions	7.G.A.2
FORWARD 2	Draw images of polygons under transformations	8.G.A.3

SET 63 • 6.G.A.3

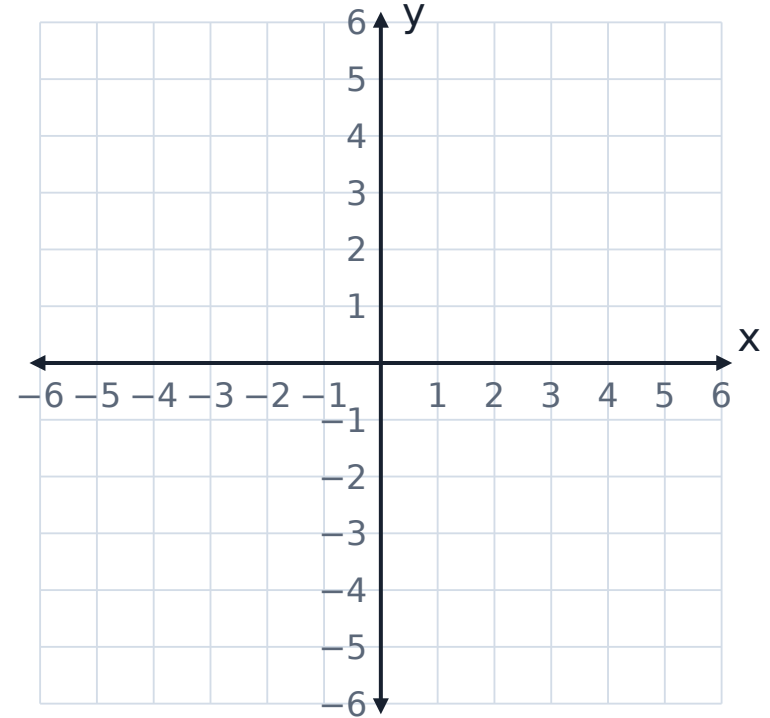
MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.3

Plot $A(-3, 2)$, $B(4, 2)$, $C(4, -3)$, and $D(-3, -3)$. Connect them in order.

What polygon did you draw?

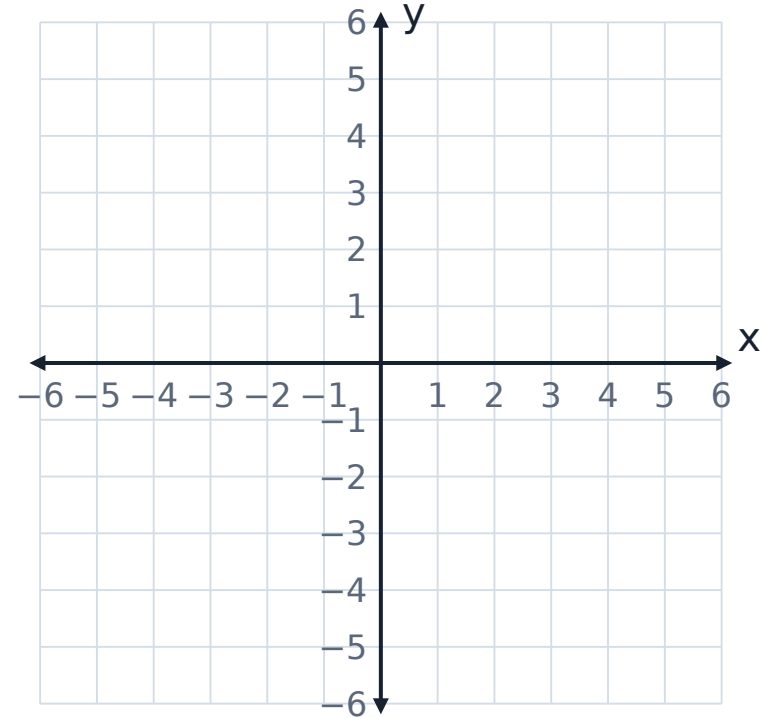
Polygon: _____



Plot $P(0, 4)$, $Q(3, -2)$, and $R(-3, -2)$. Connect them in order.

What polygon did you draw?

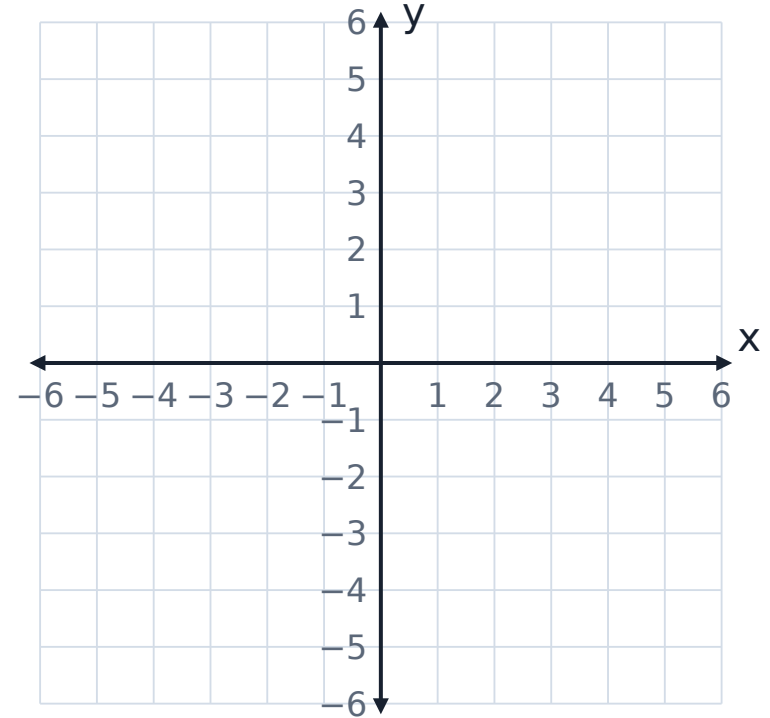
Polygon: _____



Plot $E(-2, 1)$, $F(2, 1)$, $G(4, -3)$, and $H(-4, -3)$. Connect them in order.

What polygon did you draw?

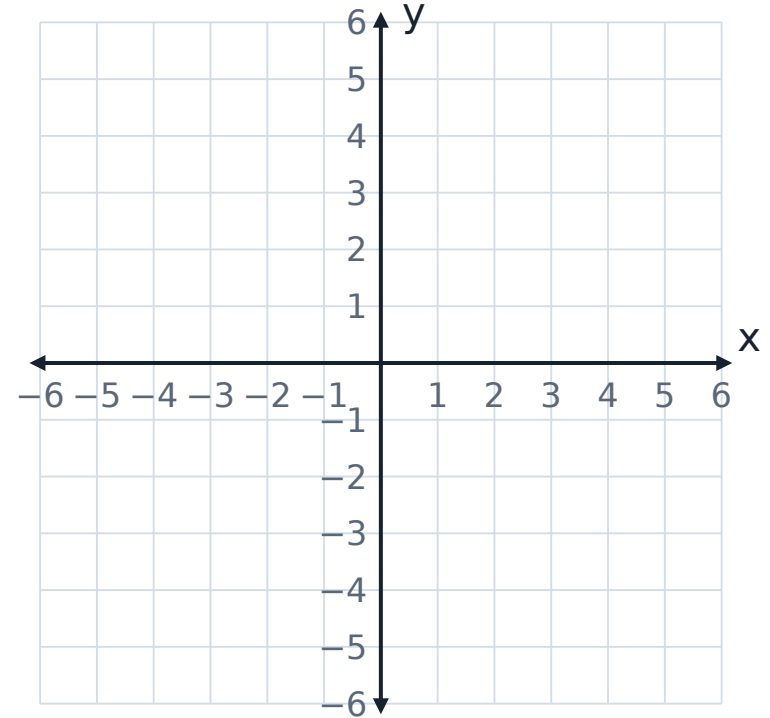
Polygon: _____



Plot $J(-1, 3)$, $K(3, 3)$, $L(3, -1)$,
and $M(-1, -1)$. Connect
them in order.

What polygon did you draw?

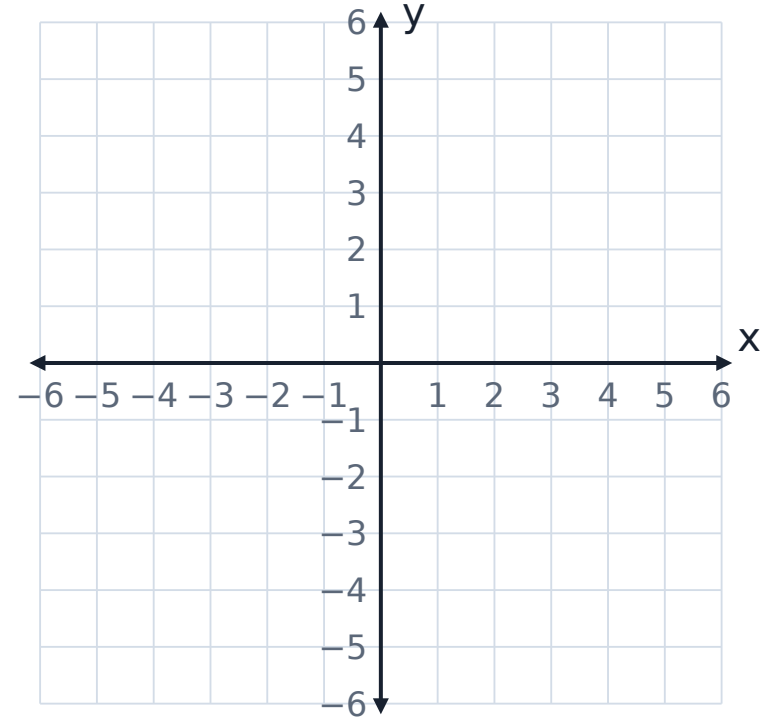
Polygon: _____



Plot $W(-5, -2)$, $X(-2, 3)$, $Y(4, 3)$, and $Z(1, -2)$. Connect them in order.

What polygon did you draw?

Polygon: _____

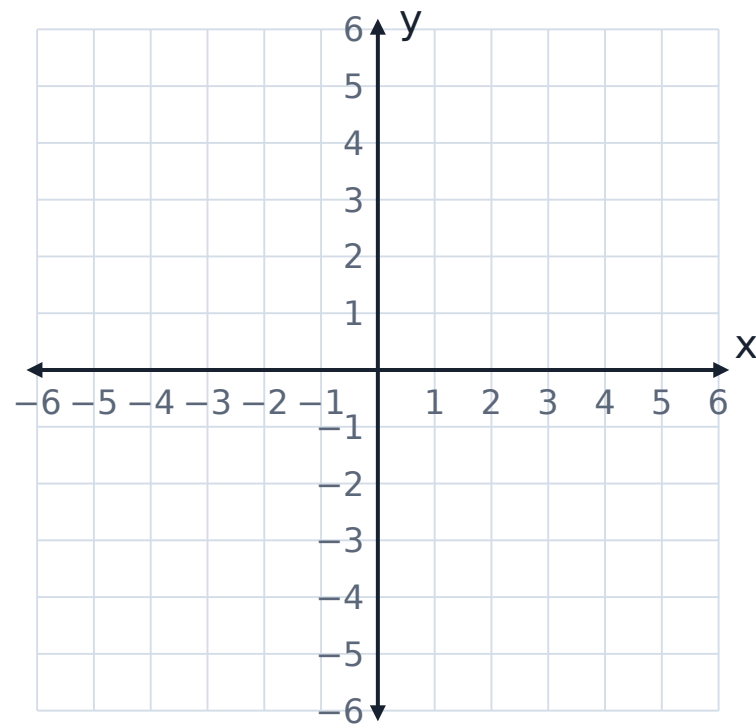


SET 63 • 6.G.A.3

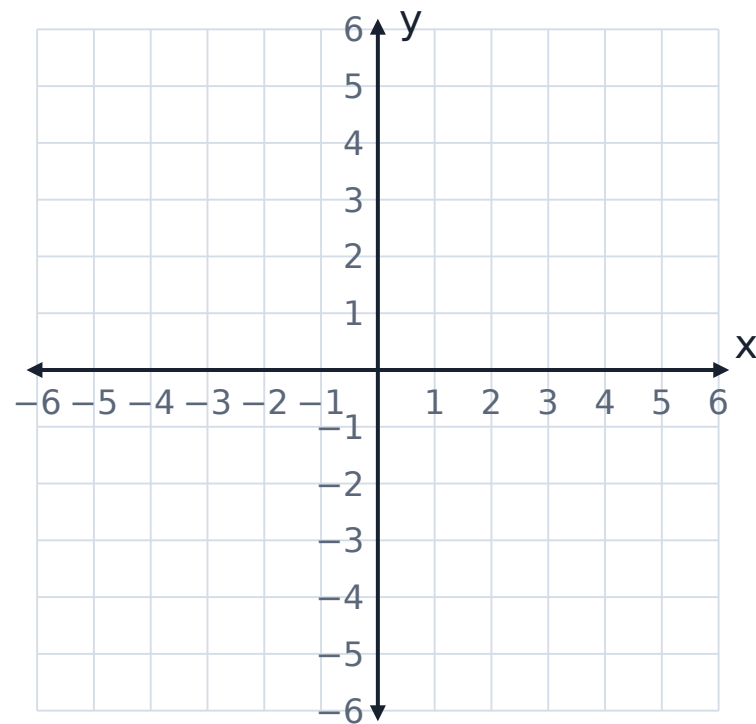
BACKWARD 1

Plotting points in all four quadrants • Grade 6 • The Number System •
6.NS.C.6.c

Plot and label point $A(-3, 2)$.



Plot and label point B(4, -3).



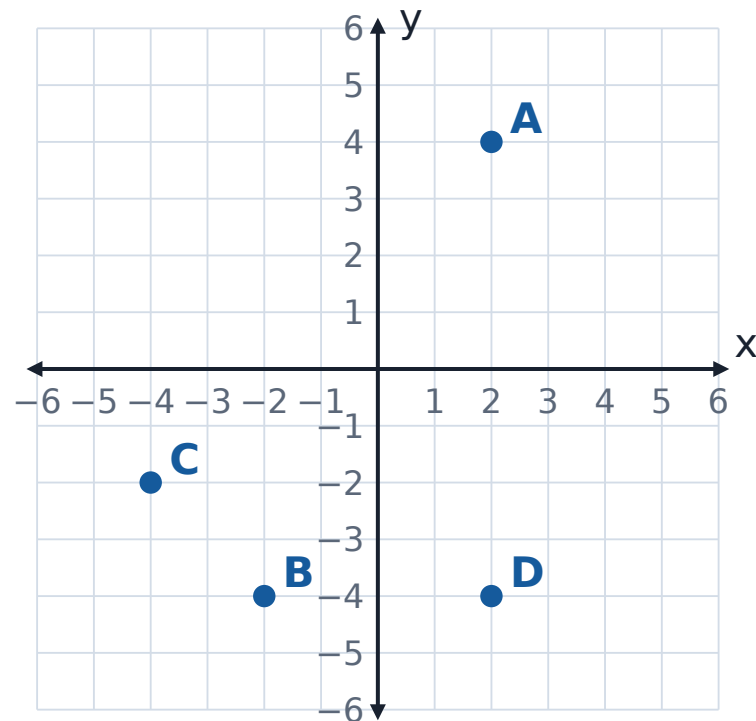
Which point is at $(-2, -4)$?

A. Point A

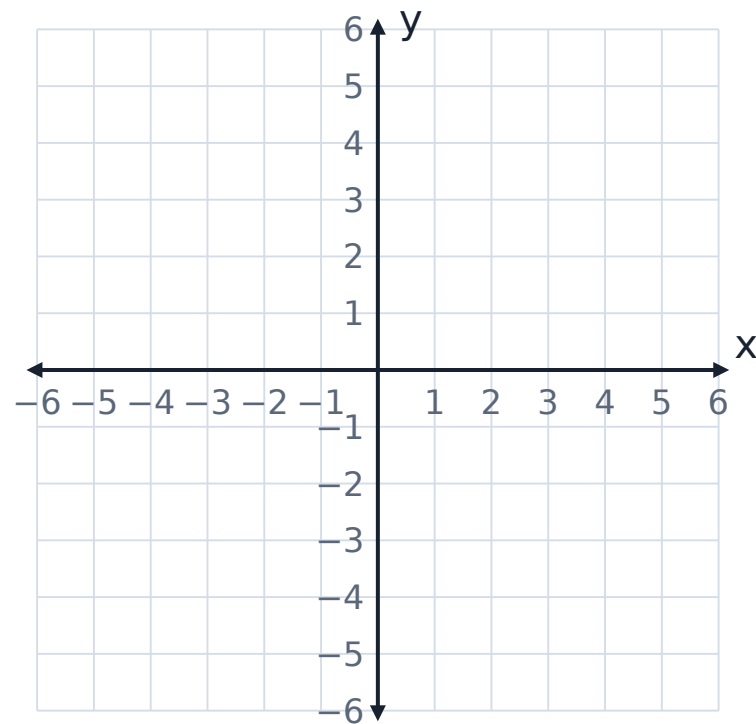
B. Point B

C. Point C

D. Point D



Plot and label point C(0, 4).

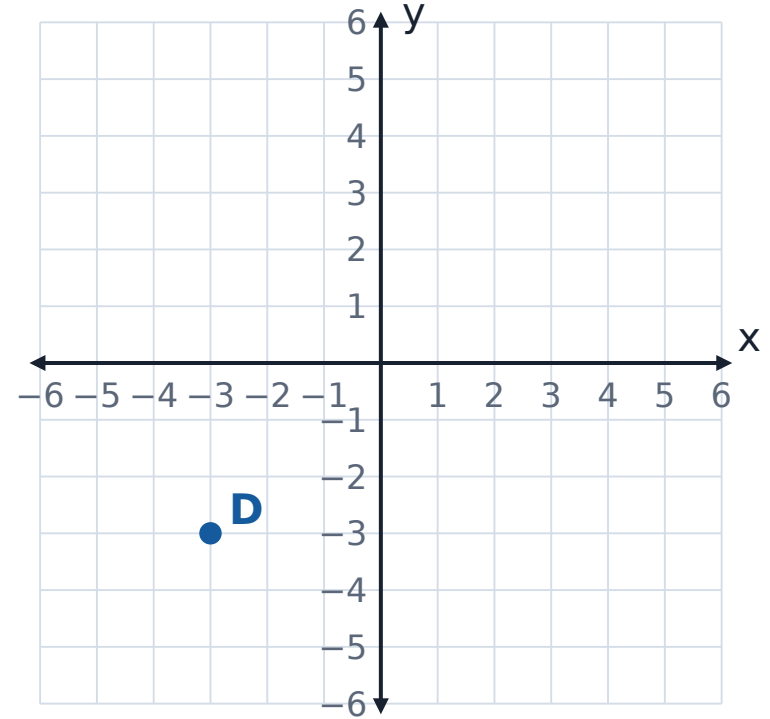


True or false?

Point D is at $(-3, -3)$.

A. True

B. False



SET 63 • 6.G.A.3

BACKWARD 2

Attributes of quadrilaterals • Grade 3 • Geometry • 3.G.A.1

True or false?

A rectangle has 4 right angles.

A. True

B. False

True or false?

Opposite sides of a rectangle have the same length.

A. True

B. False

Which shape always has 4 sides of equal length and 4 right angles?

- A.** Square
- B.** Rectangle
- C.** Trapezoid
- D.** Rhombus

True or false?

Every quadrilateral has 4 right angles.

A. True

B. False

How many sides does a quadrilateral have?

Sides: _____

SET 63 • 6.G.A.3

BACKWARD 3

Naming shapes by sides and angles • Grade 2 • Geometry • 2.G.A.1

A shape has 3 sides and 3 angles.

What is it called?

Shape: _____

How many sides does a pentagon have?

A. 4

B. 5

C. 6

D. 8

True or false?

A hexagon has 6 sides.

A. True

B. False

A shape has 4 sides and 4 angles.

What is it called?

Shape: _____

Which shape has the most sides?

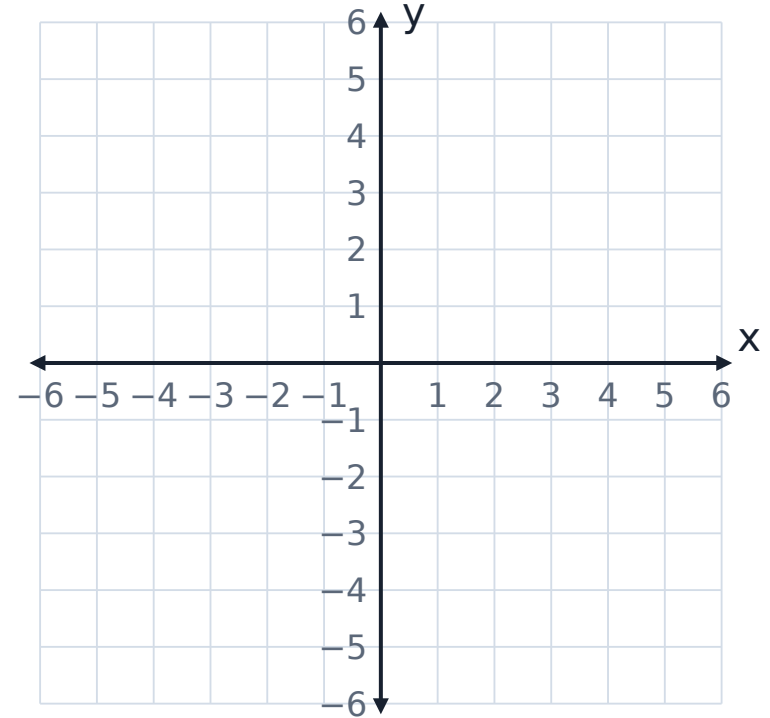
- A.** Pentagon
- B.** Quadrilateral
- C.** Triangle
- D.** Hexagon

SET 63 • 6.G.A.3

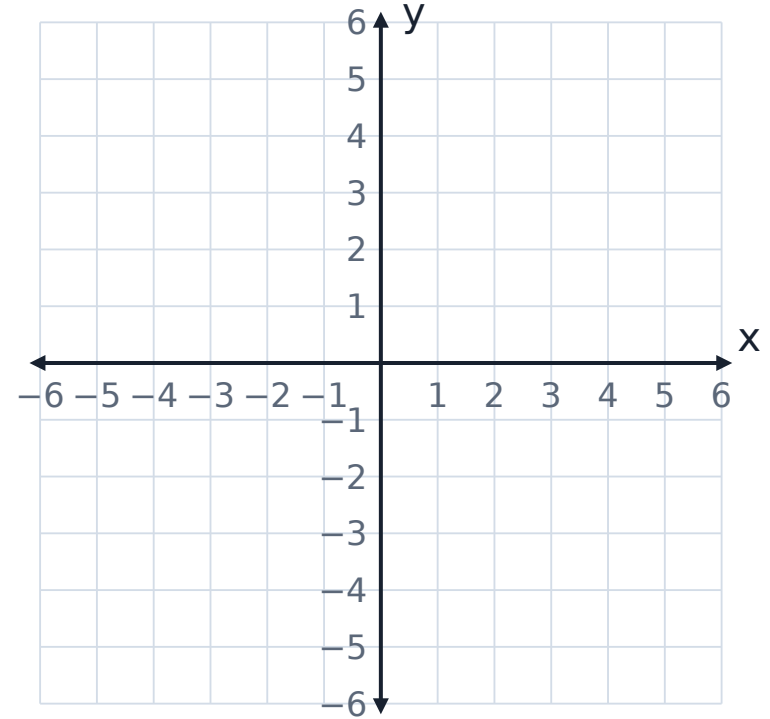
FORWARD 1

Draw shapes with given conditions • Grade 7 • Geometry • 7.G.A.2

Draw a rectangle with an area of 12 square units. One vertex is at the origin.



Draw a right triangle with legs of 4 units and 3 units. One vertex is at $(-2, -2)$.



Can a triangle have side lengths of 2 cm, 3 cm, and 6 cm? Write yes or no.

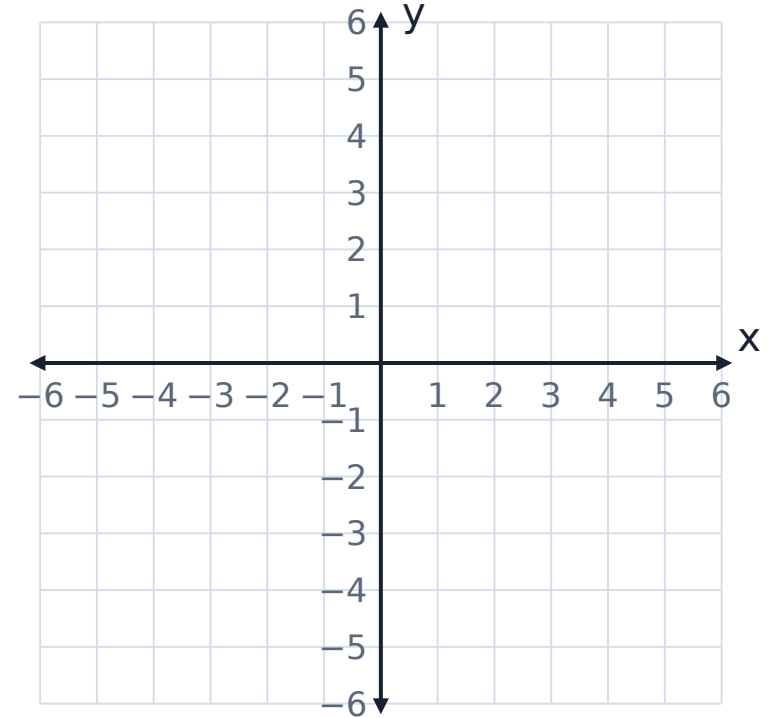
Answer: _____

A triangle has angles of 90° and 45° .

What is the third angle?

- A.** 45°
- B.** 55°
- C.** 90°
- D.** 135°

Draw a square with a side length of 5 units. One vertex is at $(-3, 1)$.



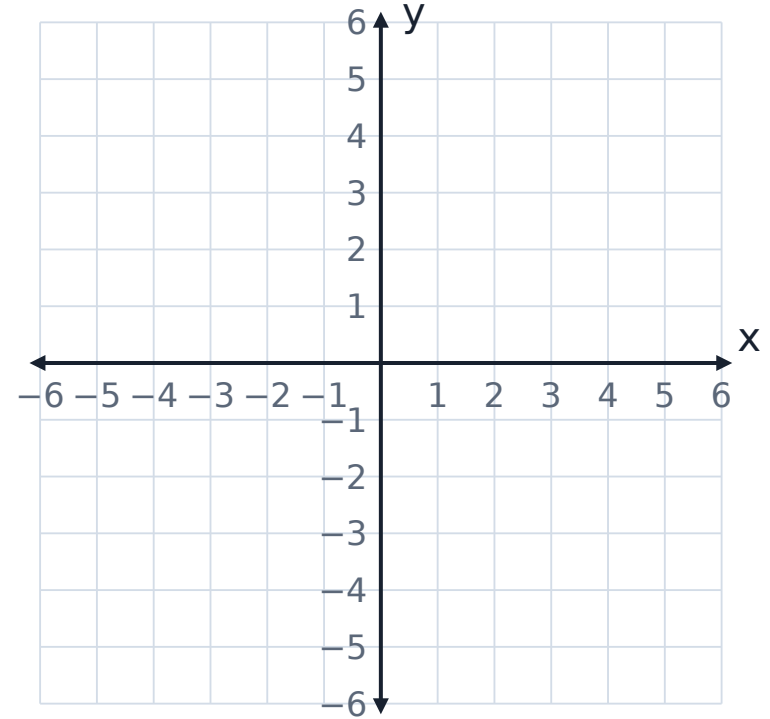
SET 63 • 6.G.A.3

FORWARD 2

Draw images of polygons under transformations • Grade 8 • Geometry •
8.G.A.3

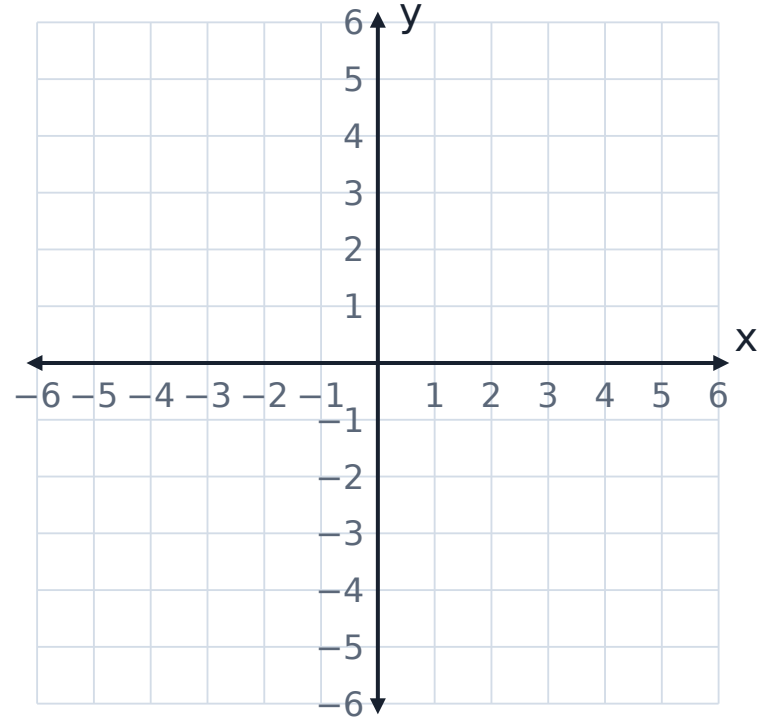
Triangle ABC has vertices
 $A(1, 1)$, $B(4, 1)$, and $C(1, 3)$.

Draw its image after a
translation 5 units left and 2
units down.



Rectangle PQRS has vertices $P(1, 2)$, $Q(4, 2)$, $R(4, 4)$, and $S(1, 4)$.

Draw its image after a reflection across the x-axis.



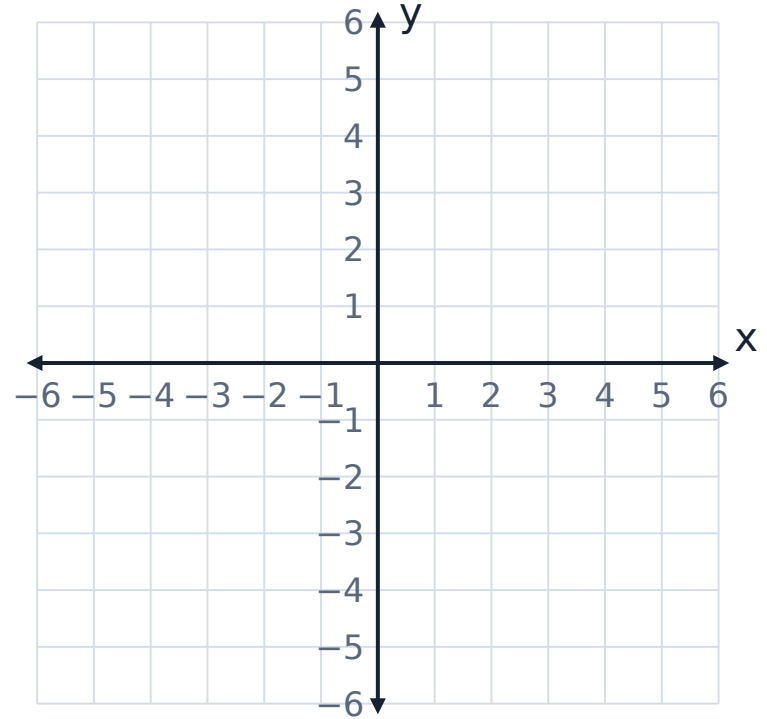
Triangle ABC has vertices A(2, 1), B(4, 3), and C(2, 3). It is rotated 180° about the origin.

What are the coordinates of B'?

B' = _____

Square EFGH has vertices $E(1, 1)$, $F(2, 1)$, $G(2, 2)$, and $H(1, 2)$.

Draw its image after a dilation by a scale factor of 3 centered at the origin.



True or false?

A rectangle with vertices $(1, 1)$, $(3, 1)$, $(3, 2)$, $(1, 2)$ is reflected across the y -axis. One image vertex is $(-3, 2)$.

A. True

B. False

Use coordinates to find side lengths and perimeters of polygons

MAIN	Grade 6 main question	6.G.A.3
BACKWARD 1	Distance between points with the same x or y	6.NS.C.8
BACKWARD 2	Absolute value	6.NS.C.7.c
BACKWARD 3	Perimeter of polygons	3.MD.D.8
FORWARD 1	Lengths between points with rational coordinates	7.NS.A.1.c
FORWARD 2	Find slanted side lengths with the Pythagorean Theorem	8.G.B.8

SET 64 • 6.G.A.3

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.3

A rectangle has vertices at $(-3, 2)$, $(4, 2)$, $(4, -5)$, and $(-3, -5)$.

What is its perimeter?

Perimeter: _____

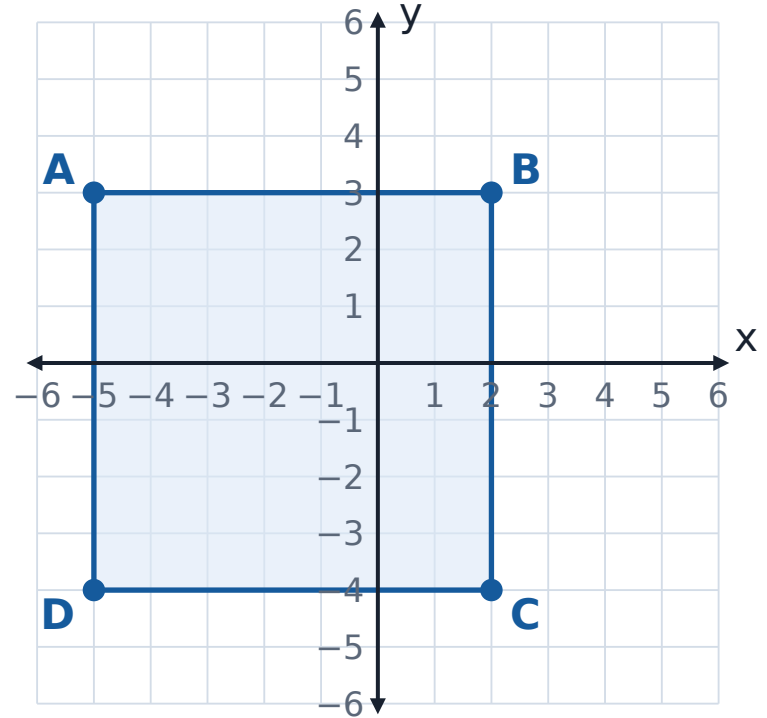
One side of a polygon goes from $(-2, -3)$ to $(-2, 5)$.

How long is this side?

Length: _____

What is the length of side AB?

AB = _____



A triangle has vertices at $(-4, 1)$, $(2, 1)$, and $(2, -7)$.

How long is the horizontal side?

- A.** 2 units
- B.** 6 units
- C.** 8 units
- D.** 10 units

A rectangle has vertices at $(-5, -2)$, $(1, -2)$, $(1, 3)$, and $(-5, 3)$.

What is its perimeter?

Perimeter: _____

SET 64 • 6.G.A.3

BACKWARD 1

Distance between points with the same x or y • Grade 6 • The Number System
• 6.NS.C.8

Find the distance between $(-3, 2)$ and $(4, 2)$.

Distance: _____

Find the distance between $(1, -4)$ and $(1, 6)$.

Distance: _____

True or false?

The distance between $(-5, 0)$ and $(5, 0)$ is 10.

A. True

B. False

What is the distance between $(2, -3)$ and $(2, -8)$?

A. -11

B. -5

C. 5

D. 11

Find the distance between $(-6, -1)$ and $(-2, -1)$.

Distance: _____

SET 64 • 6.G.A.3

BACKWARD 2

Absolute value • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$$|-5| + |2|$$

Value: _____

True or false?

$$|-7| = 7$$

A. True

B. False

$$|-3| + |4| = ?$$

A. -7

B. -1

C. 1

D. 7

Find the value.

$$|-6|$$

Value: _____

True or false?

$$|-2| + |-2| = 0$$

A. True

B. False

SET 64 • 6.G.A.3

BACKWARD 3

Perimeter of polygons • Grade 3 • Measurement & Data • 3.MD.D.8

Find the perimeter of the rectangle.

Perimeter: _____



A square has sides of 9 inches.

What is its perimeter?

Perimeter: _____

True or false?

A rectangle that is 10 m by 4 m has a perimeter of 28 m.

A. True

B. False

A triangle has sides of 5 cm, 6 cm, and 7 cm.

What is its perimeter?

- A.** 11 cm
- B.** 18 cm
- C.** 30 cm
- D.** 210 cm

A rectangle is 12 ft long and 5 ft wide.

What is its perimeter?

Perimeter: _____

SET 64 • 6.G.A.3

FORWARD 1

Lengths between points with rational coordinates • Grade 7 • The Number System • 7.NS.A.1.c

A side of a polygon goes from $(-2.5, 3)$ to $(4, 3)$.

How long is the side?

Length: _____

A rectangle has vertices at $(-1.5, 2)$, $(3.5, 2)$, $(3.5, -1)$, and $(-1.5, -1)$.

What is its perimeter?

Perimeter: _____

How long is the segment from $(0, -3\frac{1}{2})$ to $(0, 1\frac{1}{4})$?

A. $-4\frac{3}{4}$

B. $2\frac{1}{4}$

C. $4\frac{3}{4}$

D. $5\frac{1}{4}$

A square has vertices at $(-2.5, -2.5)$, $(2.5, -2.5)$, $(2.5, 2.5)$, and $(-2.5, 2.5)$.

What is the length of one side?

Length: _____

True or false?

The segment from $(-4.2, 1)$ to $(1.8, 1)$ is 6 units long.

- A. True
- B. False

SET 64 • 6.G.A.3

FORWARD 2

Find slanted side lengths with the Pythagorean Theorem • Grade 8 • Geometry
• 8.G.B.8

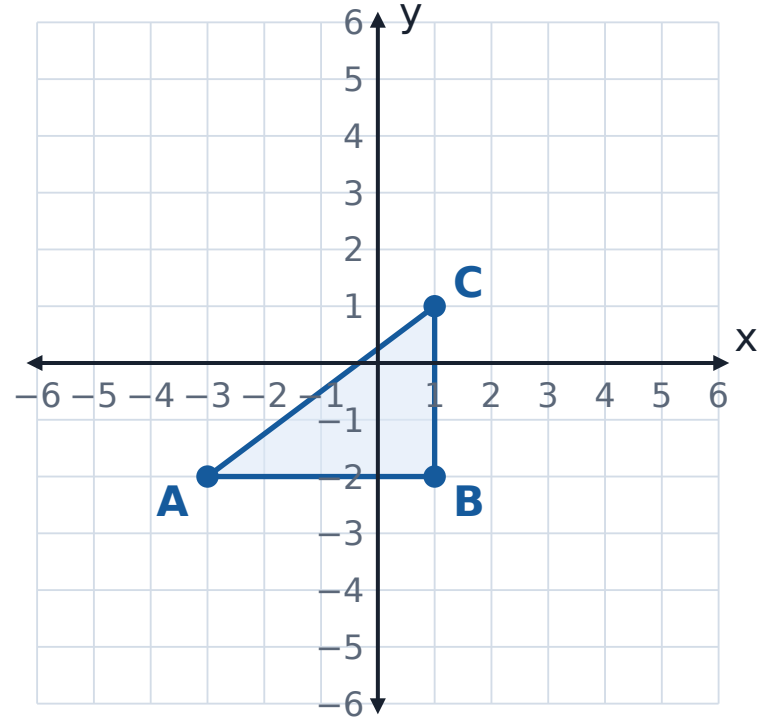
A triangle has vertices at $(0, 0)$, $(6, 0)$, and $(6, 8)$.

How long is the slanted side?

Length: _____

Find the length of side AC.

AC = _____



A parallelogram has a side from $(1, 1)$ to $(4, 5)$.

How long is that side?

- A.** 3
- B.** 5
- C.** 7
- D.** 25

Find the perimeter of the triangle with vertices $(0, 0)$, $(5, 0)$, and $(0, 12)$.

Perimeter: _____

True or false?

The side from $(-2, 3)$ to $(4, -5)$ is 10 units long.

A. True

B. False

Represent three-dimensional figures using nets

MAIN	Grade 6 main question	6.G.A.4
BACKWARD 1	Faces of three-dimensional shapes	2.G.A.1
BACKWARD 2	Shapes of faces	3.G.A.1
FORWARD 1	Cross sections of three-dimensional figures	7.G.A.3
FORWARD 2	Volume of a cylinder described by its net	8.G.C.9

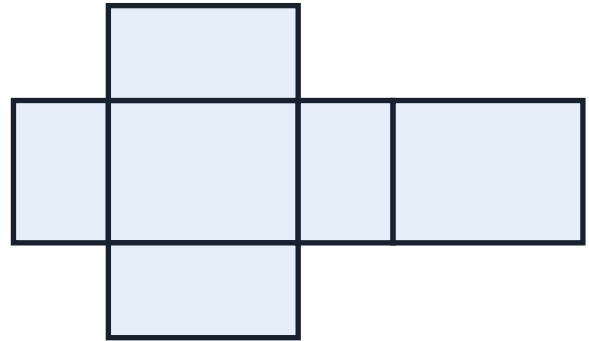
SET 65 • 6.G.A.4

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.4

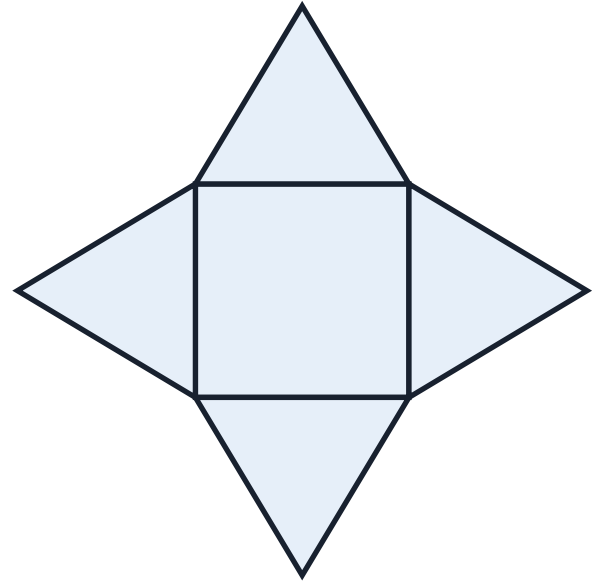
What three-dimensional figure does the net form?

- A.** Triangular prism
- B.** Rectangular prism
- C.** Square pyramid
- D.** Cube



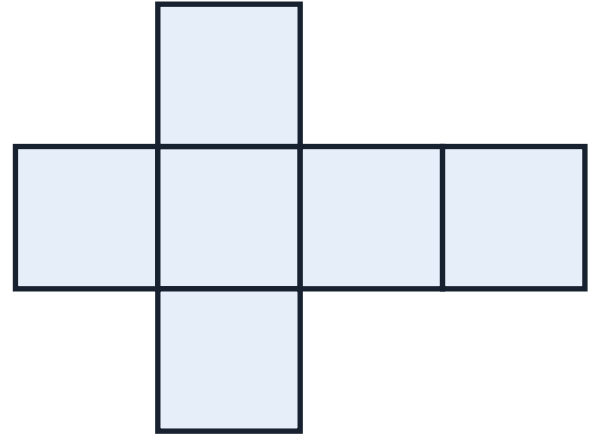
What three-dimensional figure does the net form?

- A.** Cube
- B.** Triangular pyramid
- C.** Square pyramid
- D.** Triangular prism



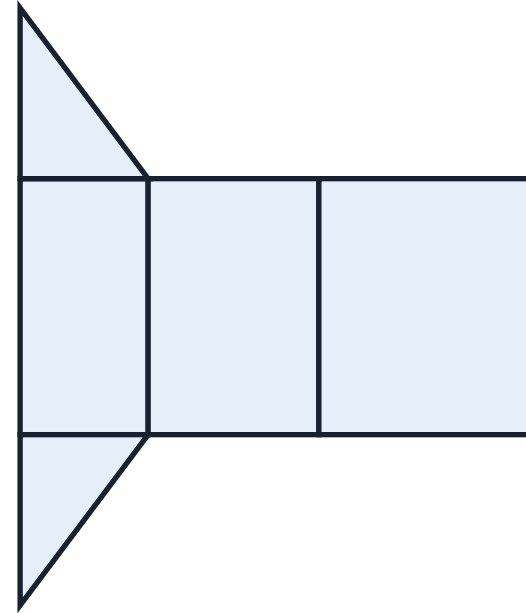
What three-dimensional figure does the net form?

- A.** Rectangular prism
- B.** Triangular prism
- C.** Cube
- D.** Square pyramid



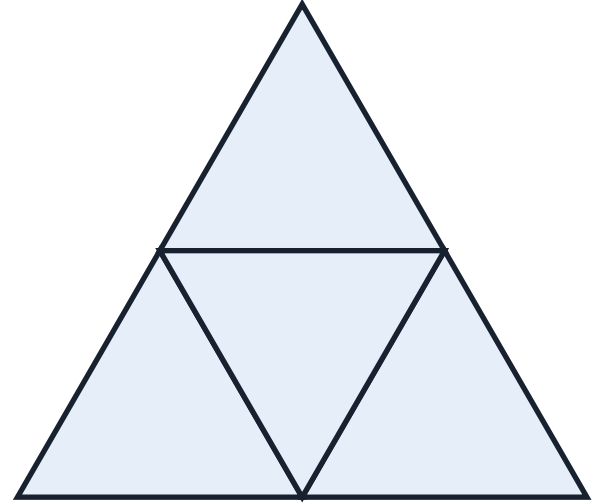
What three-dimensional figure does the net form?

- A.** Triangular prism
- B.** Triangular pyramid
- C.** Square pyramid
- D.** Rectangular prism



What three-dimensional figure does the net form?

- A.** Cube
- B.** Square pyramid
- C.** Triangular pyramid
- D.** Triangular prism



SET 65 • 6.G.A.4

BACKWARD 1

Faces of three-dimensional shapes • Grade 2 • Geometry • 2.G.A.1